

ETA6963 开关充电管理芯片数据手册

中文工程翻译版

源文件：ETA6963_V1.7.pdf

翻译原则：型号、信号名、寄存器位、单位、公式、器件料号及标准编号保持原文；说明性英文译为中文。每个原文页面后附原页图示，便于核对曲线、封装图、时序图和复杂表格。

注意：本文件为工程参考翻译，不替代芯片/器件厂商发布的最新官方英文数据手册。设计定版时应以官方原始资料为准。

原文第 1 页 - 中文译文

ETA6963

I2C 控制的 13.5V/3A 单节锂电池充电器，集成电源路径管理与 USB 充电器检测

【说明】

ETA6963 是一款高度集成的 3A 开关模式电池充电和系统电源路径管理器件，适用于单节锂离子/锂聚合物电池。它支持高输入电压和快速充电，面向智能手机、平板及各类便携设备。低阻抗电源路径可优化开关模式效率、缩短充电时间并延长电池寿命。输入电压和输入电流调节功能有助于获得最大充电功率。

器件高度集成了输入反向阻断 FET、高侧/低侧开关 FET 和电池 FET，并集成高侧栅极驱动所需的自举二极管。I2C 串行接口可配置充电和系统参数，使方案具备很高的灵活性。器件支持多种输入源，符合 USB 2.0/USB 3.0 供电规范，并提供输入电流和电压调节。器件还支持 USB OTG 工作，可在 BUS 引脚输出 5.15V，并提供最高约 1.2A 的恒流限流。

电源路径管理将系统电压调节在略高于电池电压的位置，同时不会低于可编程的 3.5V 最小系统电压。因此即使电池完全耗尽或被移除，系统仍可继续工作。当达到输入电流限制或输入电压限制时，电源路径管理会自动将充电电流降至 0；若系统负载继续增加，则由电池补充供电，直至满足系统功率需求。该补充模式（Supplement Mode）可防止输入电源过载。

ETA6963 采用 QFN 4mm×4mm、24 引脚封装。

【特性】

- 高效率 1.5MHz 同步降压充电器
- 5V 输入、2A 充电时效率约 90%
- 轻载条件下可编程 PFM 模式
- 支持 USB On-The-Go (OTG)
- 可编程限流升压转换器，输出最高 1.2A
- 宽范围单输入，可同时适配 USB 输入和高压适配器
- 输入范围 3.9V~13.5V，绝对最大输入电压 22V
- 可编程输入限流 100mA~3.2A，步进 100mA
- 自动检测 USB BC1.2、SDP、CDP、DCP 及非标准适配器
- 窄 VDC (NVDC) 电源路径管理
- BATFET 控制，支持运输模式、唤醒和整机复位
- 支持灵活的自主模式与 I2C 主机模式
- 高集成度：集成全部 MOSFET、电流检测及环路补偿
- 安全功能
- 充电/升压模式电池温度检测
- 热调节与热关断
- 输入 UVLO 欠压锁定与过压保护

【应用】

平板电脑、智能手机、互联网设备、便携音箱、手持电脑、PDA、POS 等。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



ETA6963

I²C Controlled 13.5V/3A, 1-Cell Battery Charger with Power Path Management and USB Charger Detection

DESCRIPTION

ETA6963 is a highly-integrated 3A switch-mode battery charge and system power path management device for single cell Li-Ion and Li-polymer battery. It features fast charging with high input voltage support for a wide range of smart phones, tablets and portable devices. Its low impedance power path optimizes switch-mode operation efficiency, reduces battery charging time and extends battery life. Its input voltage and current regulation deliver maximum charging power. The solution is highly integrated with input reverse-blocking FET, high/low-side switching FET, and battery FET. It also integrates the bootstrap diode for the high-side gate drive. The I²C serial interface with charging and system settings makes the device a truly flexible solution. The device supports a wide range of input sources. The device is compliant with USB 2.0 and USB 3.0 power spec with input current and voltage regulation. The device also meets USB On-the-Go (OTG) operation power rating specification by supplying 5.15 V on BUS pin with constant current limit up to 1.2A. The power path management regulates the system slightly above battery voltage but does not drop below 3.5 V minimum system voltage (programmable). With this feature, the system maintains operation even when the battery is completely depleted or removed. When the input current limit or voltage limit is reached, the power path management automatically reduces the charge current to zero. As the system load continues to increase, the power path discharges the battery until the system power requirement is met. This Supplement Mode prevents overloading the input source.

The ETA6963 is available in a QFN4x4-24L package.

FEATURES

- High-Efficiency, 1.5MHz, Synchronous Switching Buck Charger
 - 90% Charge Efficiency at 2A from 5V Input
 - Programmable PFM Mode for Light Load Conditions
- Supports USB On-The-Go (OTG)
 - Programmable Current Limit Boost Converter with Up to 1.2A Output
- Wide Range Single Input to Support both USB Input and High Voltage Adapters
 - Support 3.9V to 13.5V Input Voltage Range With 22V Absolute Maximum Input Voltage Rating
 - Programmable Input Current Limit (100mA to 3.2A With 100mA Resolution)
 - Auto Detect USB BC1.2, SDP, CDP, DCP and Non-Standard Adaptors
- Narrow VDC (NVDC) Power Path Management
- BATFET Control to Support Sleep Mode, Wake Up and Full System Reset
- Flexible Autonomous and I²C Mode for Optimal System Performance
- High Integration Includes All MOSFETs, Current Sensing and Loop Compensation
- Safety
 - Battery Temperature Sensing for Charge and Boost Mode
 - Thermal Regulation and Thermal Shutdown
- Input UVLO and Overvoltage Protection

APPLICATIONS

- Tablet PC, Smart Phone, Internet Devices
- Portable Audio Speaker, Handheld Computers, PDA, POS

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【典型应用】

典型应用图给出了 AC、BUS、PMID、STAT、nINT、SDA、SCL、nCE、DM、DP、nQON、BST、SW、SYS、BAT、LDO、TS、GND 等引脚连接。信号名保持原文，具体连接和外部元件见原页图示。

【订购信息】

料号：ETA6963Q4Y

封装：QFN4×4-24

顶部丝印：ETA6963 / YWW2L

每卷数量：5000 pcs

【引脚配置】

QFN4×4-24 顶视图见原页。主要引脚：

1 AC；2 DP；3 DM；4 STAT；5 SCL；6 SDA；7 nINT；8 NC；9 nCE；10 NC；11 TS；12 nQON；13/14 BAT；15/16 SYS；17/18 GND；19/20 SW；21 BST；22 LDO；23 PMID；24 BUS。

【绝对最大额定值】

注意：超过以下限制可能损坏器件；长期暴露在绝对最大条件下可能影响可靠性。

AC、BUS、PMID 对 GND：-2V~22V

SW 对 GND：-0.3V~22V

SYS、BAT 对 GND：-0.3V~6V

BST 对 SW：-0.3V~6V

其他所有引脚对 GND：-0.3V~6V

SW、BUS、BAT、SYS 到 PGND 的电流：内部限流

工作温度：-40 °C~85 °C

存储温度：-55 °C~150 °C

QFN4×4-24 热阻： $\theta_{JA}=35^{\circ}\text{C/W}$ ， $\theta_{JC}=10^{\circ}\text{C/W}$

引脚焊接温度（10s）：260 °C

ESD HBM：2kV

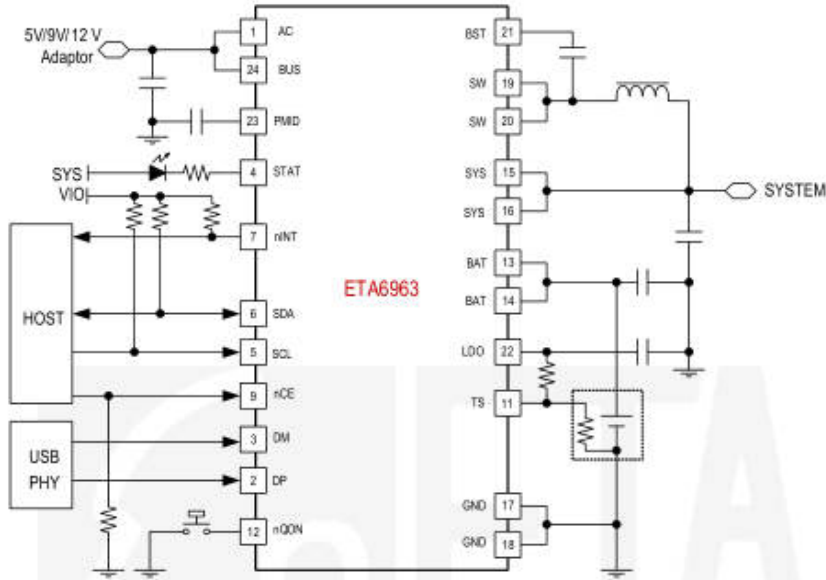
原页图示参考

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TYPICAL APPLICATION



ORDERING

PART No.
ETA6963Q4Y

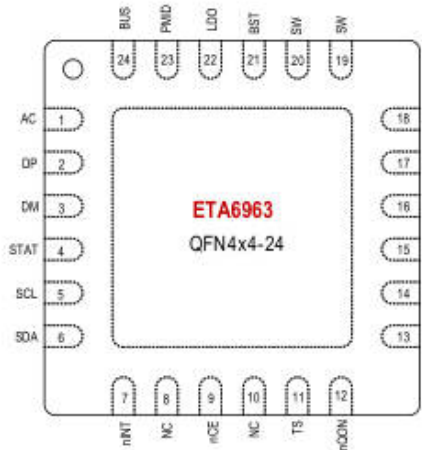
PACKAGE
QFN4x4-24

TOP MARK
ETA6963
YWW2L

Pcs/Reel
5000

INFORMATION

PIN CONFIGURATION



ABSOLUTE MAXIMUM RATINGS

(Note: Exceeding these limits may damage the device. Exposure to absolute maximum rating conditions for long periods may affect device reliability.)

- AC, BUS, PMID pin to GND Voltage-2V to 22V
- SW to GND Voltage.....-0.3V to 22V
- SYS, BAT to GND Voltage.....-0.3V to 6V
- BST to SW Voltage-0.3V to 6V
- All Other Pin to GND Voltage.....-0.3V to 6V
- SW, BUS, BAT, SYS to PGND current.....Internally limited
- Operating Temperature Range.....-40°C to 85°C
- Storage Temperature Range.....-55°C to 150°C
- Thermal Resistance θ_{JA} θ_{JC}
- QFN4x4-24.....35.....10..... °C/W
- Lead Temperature (Soldering, 10sec)260°C
- ESD HBM (Human Body Mode)2KV

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【电气特性】

条件：VBAT=3.6V，除非另有说明；典型值在 TA=25 °C 下测得。

【静态电流】

Buck 模式电池放电电流（BAT/SW/SYS）：VBAT=4.5V、VBUS<VBUS-UVLOZ、BAT 与 BUS 间漏电、TJ<85 °C，最大 5uA。

Buck 模式 BAT 放电电流：VBAT=4.5V，HIZ 模式且 BATFET_DIS=1，或无 VBUS、I2C 禁用、BATFET 关闭，TJ<85 °C；典型 20uA，最大 40uA。

BAT/SW/SYS 放电电流：VBAT=4.5V，HIZ 且 OVPFET_DIS=1，或无 VBUS、I2C 禁用、BATFET 使能，TJ<85 °C；典型 80uA，最大 160uA。

Buck 模式 VBUS 输入供电电流：VBUS>VBUS_UVLO、VBUS>VBAT、转换器开关、VBAT=3.8V、ISYS=0A；典型 5mA。

Boost 模式电池放电电流：VBAT=4.2V、Boost 模式、IBUS=0A、转换器开关；典型 2mA。

【BUS/AC/BAT 上电参数】

VBUS 工作范围（VBUS 上升）：3.9V~13.5V。

无电池时 I2C 激活的 BUS 电压：VBUS 上升，典型 3.3V，最大 3.6V。

I2C 激活迟滞：VBUS 从高于 VBUS_UVLO 下降，典型 300mV。

LDO 导通条件之一，VBUS 上升：典型 3.65V，最大 3.9V。

LDO 导通条件之一，VBUS 下降迟滞：典型 500mV。

睡眠模式下降阈值：(VBUS-VBAT)，VREG 范围内，VBUS 下降；最小 15mV、典型 60mV、最大 110mV。

睡眠模式上升阈值：最小 110mV、典型 220mV、最大 330mV。

VBUS 6.5V OVP 上升阈值（OVP=01）：6/6.5/7V（最小/典型/最大）。

VBUS 11V OVP 上升阈值（OVP=10）：10/11/12V。

VBUS 14V OVP 上升阈值（OVP=11）：13/14/15V。

VBUS 6.5V OVP 下降迟滞：典型 220mV。

VBUS 11V OVP 下降迟滞：典型 400mV。

原页图示参考

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ELECTRICAL CHARACTERISTICS

($V_{BAT} = 3.6V$, unless otherwise specified. Typical values are at $T_A = 25^\circ C$.)

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
QUIESCENT CURRENTS					
Battery discharge current (BAT, SW, SYS) in buck mode	$V_{BAT} = 4.5V$, $V_{BUS} < V_{BUS_UVLO}$, leakage between BAT and BUS, $T_J < 85^\circ C$			5	μA
Battery discharge current (BAT) in buck mode	$V_{BAT} = 4.5V$, HiZ Mode and BATFET_DIS = 1 or No V_{BUS} , I ² C disabled, BATFET Disabled, $T_J < 85^\circ C$		20	40	μA
Battery discharge current (BAT, SW, SYS)	$V_{BAT} = 4.5V$, HiZ Mode and OVPFET_DIS = 1 or No V_{BUS} , I ² C Disabled, BATFET Enabled, $T_J < 85^\circ C$		80	160	μA
Input supply current (V_{BUS}) in buck mode	$V_{BUS} > V_{BUS_UVLO}$, $V_{BUS} > V_{BAT}$, converter switching, $V_{BAT} = 3.8V$, $I_{SYS} = 0A$		5		mA
Battery Discharge Current in boost mode	$V_{BAT} = 4.2V$, boost mode, $I_{BUS} = 0A$, converter switching		2		mA
BUS, AC AND BAT PIN POWER-UP					
V_{BUS} operating range	V_{BUS} rising	3.9		13.5	V
V_{BUS} for active I ² C, no battery Sense BUS pin voltage	V_{BUS} rising		3.3	3.6	V
I ² C active hysteresis	V_{BUS} falling from above V_{BUS_UVLO}		300		mV
One of the conditions to turn on LDO	V_{BUS} rising		3.65	3.9	V
One of the conditions to turn on LDO	V_{BUS} falling		500		mV
Sleep mode falling threshold	$(V_{BUS} - V_{BAT})$, $V_{BUSMIN_FALL} \leq V_{BAT} \leq V_{REG}$, V_{BUS} falling	15	60	110	mV
Sleep mode rising threshold	$(V_{BUS} - V_{BAT})$, $V_{BUSMIN_FALL} \leq V_{BAT} \leq V_{REG}$, V_{BUS} rising	110	220	330	mV
V_{BUS} 6.5V Overvoltage rising threshold	V_{BUS} rising, OVP<1:0> = '01'	6	6.5	7	V
V_{BUS} 11V Overvoltage rising threshold	V_{BUS} rising, OVP<1:0> = '10'	10	11	12	V
V_{BUS} 14V Overvoltage rising threshold	V_{BUS} rising, OVP<1:0> = '11'	13	14	15	V
V_{BUS} 6.5V Overvoltage hysteresis	V_{BUS} falling, OVP<1:0> = '01'		220		mV
V_{BUS} 11V Overvoltage hysteresis	V_{BUS} falling, OVP<1:0> = '10'		400		mV

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Empower. Transcend. Achieve!

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【电气特性（续）】

VBUS 14V OVP 迟滞：VBUS 下降、OVP=11，典型 500mV。

无适配器时 BAT 激活 I2C：VBAT 上升，最小 2.5V、典型 2.7V。

电池耗尽下降阈值：2.2/2.4/2.6V。

电池耗尽上升阈值：2.35/2.58/2.8V。

电池耗尽上升迟滞：典型 180mV。

坏适配器检测下降阈值：VBUS 下降，3.7/3.9/4.1V。

坏适配器检测迟滞：典型 80mV。

坏适配器检测电流源：从 VBUS 向 GND 吸收，典型 30mA。

【电源路径】

系统调节电压：VBAT<SYS_MIN[2:0]=101 且 BATFET 禁用（REG07[5]=1），典型 3.68V。

当 ISYS=0A、VBAT>VSYS_MIN、BATFET 禁用时，系统调节电压典型约为 VBAT+70mV。

最大直流系统输出电压：ISYS=0A、VBAT=4.4V、BATFET 禁用，典型 4.47V。

Q1（VBUS-PMID）顶部反向阻断 MOSFET 导通电阻：-40 ° C ≤ TA ≤ 125 ° C，20/40/60mohm。

Q2（PMID-SW）高侧开关 MOSFET 导通电阻：VLDO=5V，35/70/105mohm。

Q3（SW-GND）低侧开关 MOSFET 导通电阻：VLDO=5V，21/43/65mohm。

Q4（SYS-BAT）BATFET 导通电阻：QFN，BAT 到 SYS，VBAT=4.2V，TJ=-40~125 ° C，13/27/40mohm。

补充模式下 BATFET 正向压降：典型 30mV。

【电池充电器】

充电电压编程范围：3.848V~4.616V。

充电电压步进：32mV。

VREG=4.2V 设置时：4.179/4.2/4.221V。

充电电压设置精度：VBAT=4.2V 时 ±0.5%。

充电电流调节范围：0~3000mA。

充电电流步进：60mA。

ICHG=240mA、VBAT=3.8V 时典型 0.24A；设置精度 ±10%。

电池 LOW 下降阈值：ICHG=240mA 时 2.6/2.8/3.0V。

原页图示参考

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PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
V _{BUS} 14V Overvoltage hysteresis	V _{BUS} falling, OVP<1:0> = '11'		500		mV
BAT for active I ² C, no adapter	V _{BAT} rising	2.5	2.7		V
Battery Depletion Threshold	V _{BAT} falling	2.2	2.4	2.6	V
Battery Depletion Threshold	V _{BAT} rising	2.35	2.58	2.8	V
Battery Depletion rising hysteresis	V _{BAT} rising		180		mV
Bad adapter detection falling threshold	V _{BUS} falling	3.7	3.9	4.1	V
Bad adapter detection hysteresis			80		mV
Bad adapter detection current source	Sink current from V _{BUS} to GND		30		mA
POWER-PATH					
System regulation voltage	V _{BAT} < SYS_MIN[2:0] = 101, BATFET Disabled (REG07[5] = 1)		3.68		V
System Regulation Voltage	I _{SYS} = 0A, V _{BAT} > V _{SYS_MIN} , BATFET disabled (REG07[5] = 1)		V _{BAT} + 70mV		
Maximum DC system voltage output	I _{SYS} = 0A, V _{BAT} = 4.4V, BATFET disabled (REG07[5] = 1)		4.47		V
Top reverse blocking MOSFET on-resistance between V _{BUS} and PMID - Q1	-40°C ≤ T _A ≤ 125°C	20	40	60	mΩ
Top switching MOSFET on-resistance between PMID and SW- Q2	V _{LDO} = 5 V, -40°C ≤ T _A ≤ 125°C	35	70	105	mΩ
Bottom switching MOSFET on-resistance between SW and GND- Q3	V _{LDO} = 5 V, -40°C ≤ T _A ≤ 125°C	21	43	65	mΩ
SYS-BAT MOSFET on-resistance – Q4	QFN package, Measured from BAT to SYS, V _{BAT} = 4.2V, T _J = -40-125°C	13	27	40	mΩ
BATFET forward voltage in supplement mode			30		mV
BATTERY CHARGER					
Charge voltage program range		3.848		4.616	V
Charge voltage step			32		mV
Charge voltage setting	V _{REG} (REG04[7:3] = 01011) = 4.2 V	4.179	4.2	4.221	V
Charge voltage setting accuracy	V _{BAT} = 4.2 V	-0.5		+0.5	%
Charge current regulation range		0		3000	mA
Charge current regulation step			60		mA
Charge current regulation setting	I _{CHG} = 240 mA, V _{BAT} = 3.8V		0.24		A
Charge current regulation accuracy	I _{CHG} = 240 mA, V _{BAT} = 3.8V	-10		+10	%
Battery LOW falling threshold	I _{CHG} = 240mA	2.6	2.8	3	V

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【电气特性（续）】

电池 LOW 上升阈值（预充转快充）：2.9/3.1/3.3V。

预充电流调节：IPRECHG[3:0]=0010，90/180/270mA。

终止电流调节：ITERM[3:0]=0010、VBAT=4.2V，90/180/270mA。

电池短路下降阈值：1.8/2.0/2.2V；上升阈值：2.05/2.25/2.45V。

电池短路电流：VBAT=1V 时典型 145mA。

低于 VBAT_REG 的再充电阈值：REG04[0]=0 时 70/120/170mV；REG04[0]=1 时 190/240/290mV。

【输入电压与电流调节】

输入电压调节限值：VINDPM[3:0]=0110（4.5V），4.32/4.5/4.68V，精度 $\pm 4\%$ 。

跟踪 VBAT 的输入电压调节：VINDPM=3.9V、VDPM_VBAT_TRACK=300mV、VBAT=4.0V 时，4.13/4.3/4.47V。

USB 输入电流调节限值：VBUS=5V、从 SW 拉取电流、IINDPM[4:0]=10111，2.16/2.4/2.64A。

系统启动阶段输入电流限制：典型 200mA。

【BAT 引脚过压保护】

电池过压上升阈值：VBAT 上升，为 VBAT_REG 的 108%。

过压迟滞下降阈值：VBAT 下降，为 VBAT_REG 的 103%。

【热调节与热关断】

结温调节阈值：TREG=1 时温升到典型 110 °C；TREG=0 时典型 90 °C。

热关断上升温度：典型 160 °C；热关断迟滞：典型 30 °C。

【JEITA 热敏电阻比较器（Buck 模式）】

T1（约 0 °C）阈值：低于该温度暂停充电；TS/VLDO 典型 73.3%，下降典型 71.5%。

T2（约 10 °C）阈值：低于此温区恢复至 ICHG/2 和 4.2V；TS/VLDO 典型 68%，下降典型 66.8%。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Battery LOW rising threshold	Pre-charge to Fast charge	2.9	3.1	3.3	V
Pre-charge current regulation	IPRECHG[3:0] = '0010'	90	180	270	mA
Termination current regulation	ITERM[3:0] = '0010', V _{BAT} = 4.2V	90	180	270	mA
Battery short voltage	V _{BAT} falling	1.8	2	2.2	V
Battery short voltage	V _{BAT} rising	2.05	2.25	2.45	V
Battery short current	V _{BAT} = 1V		145		mA
Recharge Threshold below V _{BAT_REG}	V _{BAT} falling, REG04[0] = 0	70	120	170	mV
Recharge Threshold below V _{BAT_REG}	V _{BAT} falling, REG04[0] = 1	190	240	290	mV
INPUT VOLTAGE AND CURRENT REGULATION					
Input voltage regulation limit	VINDPM[3:0] = 0110 = 4.5 V	4.32	4.5	4.68	V
Input voltage regulation accuracy		-4		4	%
Input voltage regulation limit tracking V _{BAT}	VINDPM[3:0] = 3.9V, V _{DPM_VBAT_TRACK} = 300mV, V _{BAT} = 4.0V	4.13	4.3	4.47	V
USB input current regulation limit	V _{BUS} = 5V, current pulled from SW, IINDPM [4:0] = 10111	2.16	2.4	2.64	A
Input current limit during system start-up sequence			200		mA
BAT PIN OVERVOLTAGE PROTECTION					
Battery overvoltage threshold	V _{BAT} rising, as percentage of V _{BAT_REG}		108		%
Battery overvoltage threshold Hysteresis	V _{BAT} falling, as percentage of V _{BAT_REG}		103		%
THERMAL REGULATION AND THERMAL SHUTDOWN					
Junction Temperature Regulation Threshold	Temperature Increasing, TREG = 1		110		°C
Junction Temperature Regulation Threshold	Temperature Increasing, TREG = 0		90		°C
Thermal Shutdown Rising Temperature			160		°C
Thermal Shutdown Hysteresis			30		°C
JEITA Thermistor Comparator (BUCK MODE)					
T1 (0°C) threshold, Charge suspended T1 below this temperature.	Charger suspends charge. As Percentage to V _{LDO}		73.3		% V _{LDO}
Falling	As Percentage to V _{LDO}		71.5		% V _{LDO}
T2 (10°C) threshold, Charge back to I _{CHG} /2 and 4.2 V below this temperature	As Percentage to V _{LDO}		68		% V _{LDO}
Falling	As Percentage to V _{LDO}		66.8		% V _{LDO}

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【JEITA 热敏电阻比较器（续）】

T3 (约 45 ° C) 阈值：温度高于该点时回到 ICHG 和 4.05V 相关策略；TS/VLDO 典型 44.7%，下降 45.7%。

T5 (约 60 ° C) 阈值：高于该温度暂停充电；TS/VLDO 典型 34.2%，下降 35.3%。

【冷/热热敏电阻比较器（Boost 模式）】

冷温阈值：TS 电压上升阈值，约对应 103AT 热敏电阻 -20 ° C；TJ=-20~125 ° C，78/80/82% VLDO；下降 77/79/81% VLDO。

热温阈值：TS 电压下降阈值，约对应 103AT 60 ° C；29.2/31.2/33.2% VLDO；上升 32.4/34.4/36.4% VLDO。

【逐周期充电过流比较器】

高侧 FET 逐周期过流阈值：4.8/5.8/7.5A。

系统过载阈值：4.7/5.2/5.7A。

【逐周期欠流比较器】

低侧 FET 欠流下降阈值：从同步模式切换到非同步模式，典型 250mA。

【PWM】

开关频率：典型 1500kHz。

最大 PWM 占空比：典型 97%。

【Boost 模式】

升压稳压电压：VBAT=3.8V、IPMID=0A、BOOSTV[1:0]=10，5.0/5.15/5.3V；精度 ±3%。

退出 Boost 的电池电压：

- MIN_VBAT_SEL=0：下降 2.6/2.8/3.0V，上升 2.8/3.0/3.2V；

- MIN_VBAT_SEL=1：下降 2.3/2.5/2.7V，上升 2.6/2.8/3.0V。

OTG 输出电流：BOOST_LIM=1 时典型 1.4A（规格最低 1.2A）；BOOST_LIM=0 时典型 0.7A（规格最低 0.5A）。

OTG 过压上升阈值：5.6/6.0/6.4V。

高侧 FET 欠流下降阈值：典型 100mA。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
T3 (45°C) threshold, charge back to I_{CHG} and 4.05V above this temperature.	Charger suspends charge. As Percentage to V_{LDO}		44.7		% V_{LDO}
Falling	As Percentage to V_{LDO}		45.7		% V_{LDO}
T5 (60°C) threshold, charge suspended above this temperature	As Percentage to V_{LDO}		34.2		% V_{LDO}
Falling	As Percentage to V_{LDO}		35.3		% V_{LDO}
COLD OR HOT THERMISTER COMPARATOR (BOOST MODE)					
Cold Temperature Threshold, TS pin Voltage Rising Threshold	As Percentage to V_{LDO} (Approx. -20°C w/ 103AT) $T_J = -20^{\circ}\text{C} - 125^{\circ}\text{C}$	78	80	82	% V_{LDO}
Falling	$T_J = -20^{\circ}\text{C} - 125^{\circ}\text{C}$	77	79	81	% V_{LDO}
Hot Temperature Threshold, TS pin Voltage falling Threshold	As Percentage to V_{LDO} (Approx. 60°C w/ 103AT), $T_J = -20^{\circ}\text{C} - 125^{\circ}\text{C}$	29.2	31.2	33.2	% V_{LDO}
Rising	$T_J = -20^{\circ}\text{C} - 125^{\circ}\text{C}$	32.4	34.4	36.4	% V_{LDO}
CHARGE OVERCURRENT COMPARATOR (CYCLE-BY-CYCLE)					
HSFET cycle-by-cycle over-current threshold		4.8	5.8	7.5	A
System over load threshold		4.7	5.2	5.7	A
CHARGE UNDER-CURRENT COMPARATOR (CYCLE-BY-CYCLE)					
LSFET under-current falling threshold	From sync mode to non-sync mode		250		mA
PWM					
PWM switching frequency	Oscillator frequency		1500		kHz
Maximum PWM duty cycle			97		%
BOOST MODE OPERATION					
Boost mode regulation voltage	$V_{BAT} = 3.8\text{ V}$, $I_{PMID} = 0\text{ A}$, BOOSTV[1:0] = '10'	5	5.15	5.3	V
Boost mode regulation voltage accuracy	$V_{BAT} = 3.8\text{ V}$, $I_{PMID} = 0\text{ A}$, BOOSTV[1:0] = '10'	-3		3	%
Battery voltage exiting boost mode	V_{BAT} falling, MIN_VBAT_SEL = 0	2.6	2.8	3.0	V
	V_{BAT} rising, MIN_VBAT_SEL = 0	2.8	3	3.2	V
	V_{BAT} falling, MIN_VBAT_SEL = 1	2.3	2.5	2.7	V
	V_{BAT} rising, MIN_VBAT_SEL = 1	2.6	2.8	3.0	V
OTG mode output current	BOOST_LIM = 1	1.2	1.4		A
	BOOST_LIM = 0	0.5	0.7		A
OTG overvoltage threshold	Rising threshold	5.6	6	6.4	V
HSFET under current falling threshold			100		mA

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【Boost Hiccup 与 LDO】

Hiccup 重试定时：典型 7ms。

Hiccup 关闭时间：典型 28ms。

最大允许重试次数：7 次；最后一次重试后器件清除 OTG_CONFIG 位以关闭 Boost。

LDO 输出电压：VBUS=9V、ILDO=40mA 时典型 5V。

LDO 输出限流：VBUS=9V、VLDO=4V 时典型 80mA。

【逻辑 I/O - nCE、SCL、SDA】

输入低阈值：最大 0.4V。

输入高阈值：最小 1.3V。

高电平漏电流：上拉电源 1.8V，最大 1uA。

【逻辑 I/O - STAT、nINT】

低电平输出电压：最大 0.4V。

【VBUS/BAT 上电】

坏适配器检测持续时间：典型 30ms。

【电池充电器定时】

充电终止去毛刺：典型 7s。

再充电去毛刺：典型 32ms。

系统过流导致 Q4 关闭的去毛刺：典型 1ms。

电池过压关闭充电去毛刺：典型 10us。

充电安全计时器：CHG_TIMER=1 时典型 10h。

Top-Off 定时器：TOP_OFF_TIMER[1:0]=10 时典型 30min。

【nQON 时序】

退出运输模式、打开 BATFET 的 nQON 低电平时间：-10 ° C ≤ T_J ≤ 60 ° C，典型 1.15s。

复位 BATFET 的 nQON 低电平时间：典型 10s。

整机复位期间 BATFET 关闭时间：典型 320ms。

进入运输模式延迟：典型 12s。

【数字时钟与看门狗】

REG05[4]=01 时看门狗时间：40s。

【DP/DM 检测】

VDP/DM_600MVSRC：600mV 电压源。

RDM_19K：19kohm。

VDP/DM_0P325：0.325V。

V2P7_VTH：非标准适配器 DP/DM 阈值 2.55/2.7/2.85V。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Hiccup Retrying timer			7		ms
Hiccup Off timer			28		ms
Maximum retry allowed	After the last retry, part clear OTG_CONFIG bit to disable Boost		7		Time
VLDO REGULATION					
LDO output voltage	$V_{BUS} = 9V$, $I_{LDO} = 40mA$		5		V
LDO output current limit	$V_{BUS} = 9V$, $V_{LDO} = 4V$		80		mA
LOGIC I/O PIN CHARACTERISTICS (nCE, SCL, SDA)					
Input low threshold				0.4	V
Input high threshold		1.3			V
High-level leakage current	Pull up rail 1.8 V			1	μA
LOGIC I/O PIN CHARACTERISTICS (STAT, nINT)					
Low-level output voltage				0.4	V
VBUS/BAT POWER UP					
Bad adapter detection duration			30		ms
BATTERY CHARGER					
Deglitch time for charge Termination			7		S
Deglitch time for recharge			32		ms
System over-current deglitch time to turn off Q4			1		ms
Battery over-voltage deglitch time to disable charge			10		μs
Typical Charge Safety Timer Range	CHG_TIMER = 1		10		Hr
Typical Top-Off Timer Range	TOP_OFF_TIMER[1:0] = 10		30		min
nQON TIMING					
nQON low time to turn on BATFET and exit ship mode	$-10^{\circ}C \leq T_J \leq 60^{\circ}C$		1.15		s
nQON low time to reset BATFET	$-10^{\circ}C \leq T_J \leq 60^{\circ}C$		10		s
BATFET off time during full system reset	$-10^{\circ}C \leq T_J \leq 60^{\circ}C$		320		ms
Enter ship mode delay	$-10^{\circ}C \leq T_J \leq 60^{\circ}C$		12		s
DIGITAL CLOCK AND WATCHDOG TIMER					
REG05[4]=01			40		s
DP/DM DETECTION					
VDP/DM_600MVSRC	Voltage source (600 mV)		600		mV
RDM_19K			19		k Ω
VDP/DM_0P325			0.325		V
V2P7_VTH	DP/DM Threshold for non-standard adapter	2.55	2.7	2.85	V

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【DP/DM 检测（续）】

V2P0_VTH：非标准适配器 DP/DM 阈值 1.85/2.0/2.15V。

V1P2_VTH：非标准适配器 DP/DM 阈值 1.05/1.2/1.35V。

【引脚说明】

AC (1)：充电器输入电压检测。必须与 BUS 引脚相连。

BUS (24)：充电器输入电压。BUS 到 PGND 连接 10uF 陶瓷旁路电容，并尽量靠近 BUS 引脚。

DP (2)：USB 数据线正端。用于基于 DP/DM 的 USB 主机/充电端口检测，包括数据接触检测 (DCD)、BC1.2 一次/二次检测及非标准适配器检测。

DM (3)：USB 数据线负端，功能与 DP 配合完成上述检测。

STAT (4)：开漏充电状态输出。通过约 10kohm 电阻上拉到逻辑电源，也可串限流电阻与 LED 指示状态：充电中=低；充电完成或充电器休眠=高；充电暂停（故障响应）或无电池=1Hz、50% 占空比脉冲。

SCL (5)：I2C 时钟，建议用 10kohm 上拉到逻辑电源。

SDA (6)：I2C 数据，建议用 10kohm 上拉到逻辑电源。

nINT (7)：开漏中断输出。通过 10kohm 上拉；向主机输出低有效、约 256us 脉冲，用于报告充电器状态和故障。

NC (8)：不连接，保持悬空。

nCE (9)：充电禁止控制。nCE=0 使能充电，nCE=1 禁止充电。

NC (10)：不连接，保持悬空。

TS (11)：支持 JEITA 温度策略的温度检测输入。连接 NTC 热敏电阻，并用 LDO-TS-GND 分压设置温度窗口。当 TS 超出范围时暂停充电。不使用时可从 LDO 到 TS 接 10kohm，同时 TS 到 GND 接 10kohm。推荐使用 103AT-2 热敏电阻。

nQON (12)：BATFET 使能/复位控制。BATFET 处于运输模式时，保持低电平约 tSHIPMODE（典型 1.15s）可打开 BATFET 并退出运输模式。当 VBUS 未接入时，保持低电平至少 tQON_RST（最小约 8s）可通过关闭 BATFET 至少 tBATFET_RST（约 250ms）再重新开启，实现 SYS 整机电源复位。该引脚内部带上拉，默认保持高电平。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
V2P0_VTH	DP/DM Threshold for non-standard adapter	1.85	2.0	2.15	V
V1P2_VTH	DP/DM Threshold for non-standard adapter	1.05	1.2	1.35	V

PIN DESCRIPTION

PIN NAME	PIN #	DESCRIPTION
AC	1	Charger input voltage sense. This pin must be connected to BUS pin
BUS	24	Charger input voltage. Bypass it with a 10 μ F ceramic capacitor from BUS to PGND. The capacitor should be close to the BUS pin
DP	2	Positive line of the USB data line pair. DP/DM based USB host/charging port detection. The detection includes data contact detection (DCD), primary and secondary detection in BC1.2 and nonstandard adaptors
DM	3	Negative line of the USB data line pair. DP/DM based USB host/charging port detection. The detection includes data contact detection (DCD), primary and secondary detection in BC1.2 and nonstandard adaptors
STAT	4	Open-drain charge status output. Connect the STAT pin to a logic rail via 10k Ω resistor. The STAT pin indicates charger status. Connect a current limit resistor and a LED from a rail to this pin. Charge in progress: LOW Charge complete or charger in SLEEP mode: HIGH Charge suspend (fault response) or No bat: 1Hz, 50% duty cycle Pulses
SCL	5	I ² C interface clock. Connect a 10k Ω pull up resistor to the logic rail
SDA	6	I ² C interface data. Connect a 10k Ω pull up resistor to the logic rail
nINT	7	Open-drain interrupt Output. Connect the nINT pin to a logic rail through 10k Ω resistor. The nINT pin sends an active low, 256 μ s pulse to host to report charger device status and fault
NC	8	No Connect. Keep the pin float
nCE	9	Charge disable control pin. nCE = 0, charge is enabled. nCE = 1, charge is disabled
NC	10	No Connect. Keep the pin float
TS	11	Temperature qualification voltage input to support JEITA profile. Connect a negative temperature coefficient thermistor. Program temperature window with a resistor divider from LDO to TS to GND. Charge suspends when either TS pin is out of range. When TS pin is not used, connect a 10k Ω resistor from LDO to TS and connect a 10k Ω resistor from TS to GND. It is recommended to use a 103AT-2 thermistor
nQON	12	BATFET enable/reset control input. When BATFET is in ship mode, a logic low of t _{SHIPMODE} (typical 1.15s) duration turns on BATFET to exit shipping mode. When V _{BUS} is not plugged-in, a logic low of t _{QON_RST} (minimum 8s) duration resets SYS (system power) by turning BATFET off for t _{BATFET_RST} (minimum 250ms) and then re-enable BATFET to provide full system power reset. The pin contains an internal pull-up to maintain default high logic

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原文第 9 页 - 中文译文

【引脚说明（续）】

BAT（13、14）：电池正端。内部 BATFET 和电流检测网络位于 SYS 与 BAT 之间。BAT 引脚附近连接 10uF 电容。

SYS（15、16）：转换器输出/系统供电节点。内部电流检测网络位于 SYS 与 BAT 之间。SYS 引脚附近建议连接 $2 \times 10\mu\text{F}$ 电容。

GND（17、18）：功率地。

SW（19、20）：开关节点输出，连接输出电感。BST 与 SW 之间连接 10nF 自举电容。

BST（21）：高侧 FET 栅极驱动自举电容连接端，BST-SW 之间接 10nF 陶瓷电容。

LDO（22）：LDO 输出。LDO 到 GND 连接 4.7uF、耐压 10V 电容，并尽量靠近引脚。

PMID（23）：反向阻断 FET 与高侧开关 FET 之间的节点。PMID 到 PGND 连接 $2 \times 10\mu\text{F}$ 电容，并尽量靠近引脚。

【典型性能特性】

测试典型值在 $T_A=25^\circ\text{C}$ 下获得，除非另有说明。

左图：OTG 输出电压 V_{OUT} 随输出电流 I_{OUT} 的变化，目标 $V_{\text{OUT}}=5.1\text{V}$ ，分别给出 $V_{\text{BAT}}=3.3\text{V}$ 、 3.6V 、 3.8V 、 4.2V 、 4.35V 曲线。

右图：OTG 效率随 I_{OUT} 的变化，采用相同电池电压条件。

具体曲线和坐标见原页。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

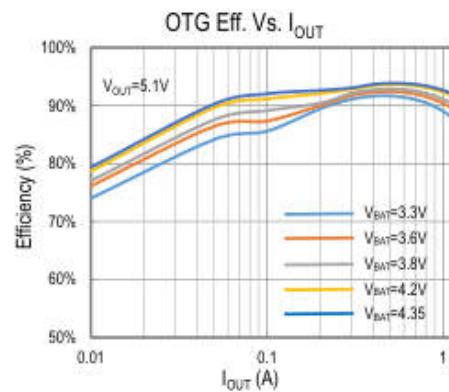
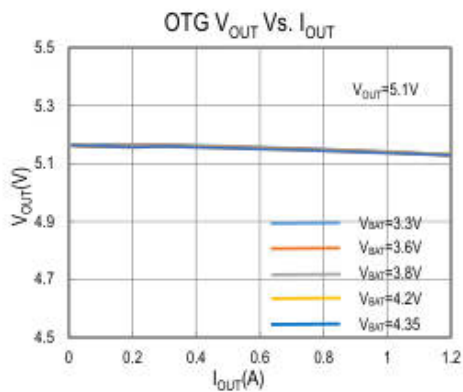


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PIN NAME	PIN #	DESCRIPTION
BAT	13, 14	Positive battery terminal. The internal BATFET and current sensing is connected between SYS and BAT. Connect a 10 μ F close to the BAT pin
SYS	15, 16	Converter output connection point. The internal current sensing network is connected between SYS and BAT pin. Connect 2 x10 μ F close to the SYS pin
GND	17, 18	Power Ground
SW	19, 20	Switching node output . Connected to output inductor. Connect the 10nF bootstrap capacitor from SW pin to BST pin
BST	21	Bootstrap capacitor connection for the high-side FET gate driver. Connect a 10nF ceramic capacitor from BST pin to SW pin
LDO	22	LDO Output Voltage. Bypass the pin with 4.7 μ F (10V rating) capacitor from LDO to GND. The capacitor should be closed to the pin
PMID	23	Connection point between reverse blocking FET and high-side switching FET. Bypass it with 2 x10 μ F capacitor from PMID to PGND. This capacitor should be close to the PMID pin

TYPICAL PERFORMANCE CHARACTERISTICS

(Typical values are at $T_A = 25^\circ\text{C}$ unless otherwise specified.)



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【典型性能特性（续）】

- 充电效率随输出/充电电流 IOUT 的变化，分别给出 VBAT=3.8V、VIN=5V/9V/12V 的曲线。
 - 充电效率随 IOUT 的变化，分别给出 VBAT=3.3V、VIN=5V/9V/12V 的曲线。
 - 充电终止电压 VTERM 随温度变化：VBUS=5V。
 - 输入电流调节 IINDPM 随温度变化：VBUS=5V，IINDPM=1A。
 - 输入电压动态功率管理阈值 VINDPM 随温度变化。
 - 充电电流 ICHG 随温度变化：VBUS=5V、VBAT=3.3V，并对比 TREG[]=1（REG05=9F）与 TREG[]=0（REG05=9D）的热调节行为。
- 具体曲线及坐标见原页。

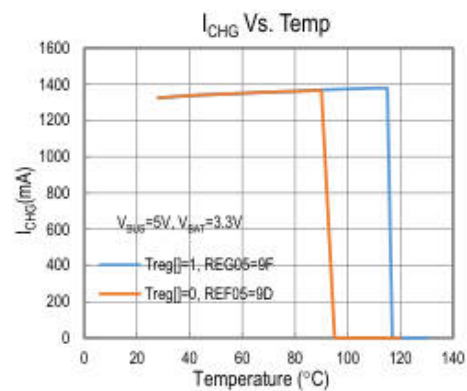
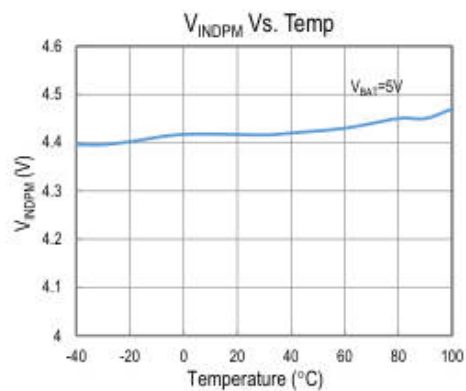
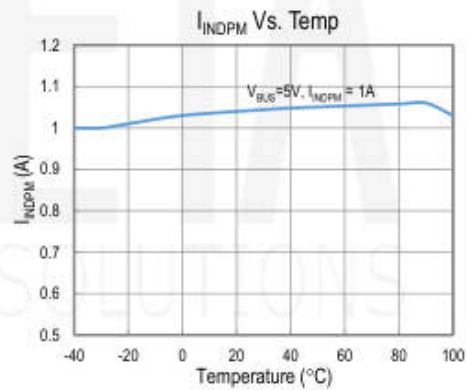
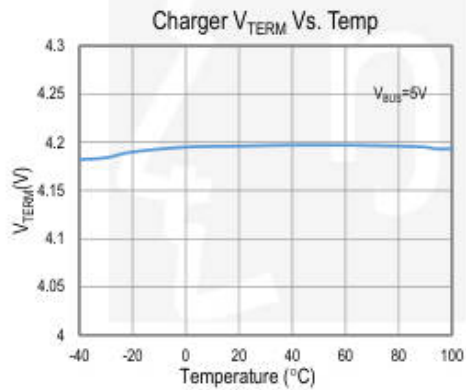
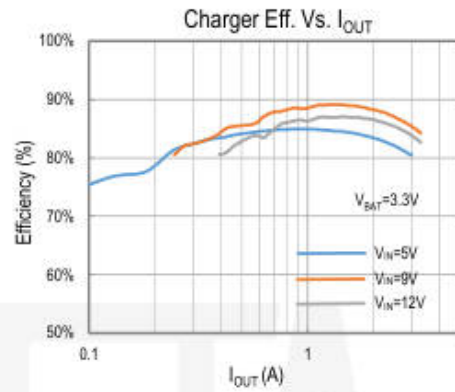
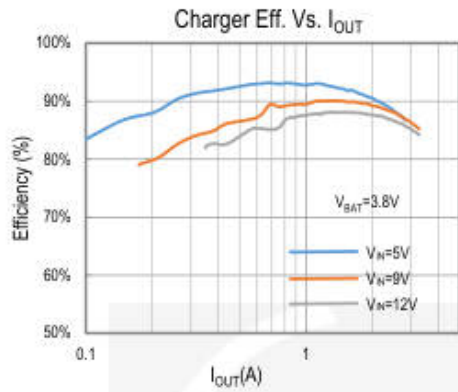
原页图示参考

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TYPICAL PERFORMANCE CHARACTERISTICS cont'


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【功能说明】

ETA6963 是一款高度集成的 3A 单节锂离子/锂聚合物电池开关模式充电器，内部集成输入反向阻断 FET（RBFET，Q1）、高侧开关 FET（HSFET，Q2）、低侧开关 FET（LSFET，Q3）、电池 FET（BATFET，Q4）以及高侧栅极驱动自举二极管。

【无输入源、仅电池上电】

仅接入电池而没有外部输入时，器件检测电池电压并打开 BATFET，但不会开启 VLDO，以尽量降低供电电流，尤其是避免 NTC 分压网络消耗。此时系统全部由电池直接供电。只有启用 BOOST 后 VLDO 才会开启。为优化轻载时的电池放电电流，BATFET 栅极电荷泵以较低频率工作。

【接入输入源上电】

接入输入电源后，器件检测输入电压并开启 LDO 及全部偏置电路。在 Buck 转换器启动前，会先检测并设置输入电流限制。输入源上电顺序如下：

1. 启动 VLDO；
2. 执行劣质电源源资格检测；
3. 基于 D+/D- 检测输入源类型，并设置默认输入限流 IINDPM 或输入源类型；
4. 设置输入电压限制阈值 VINDPM；
5. 启动转换器。

【LDO 上电】

满足以下条件时 VLDO 使能：

1. VBUS 高于 3.65V；
2. Buck 模式下 $VBUS > VBAT + VSLEEP$ （约 200mV），或 Boost 模式下 $VBUS < VBAT + VSLEEP$ 。

上述两个条件成立后，LDO 立即启动。

【电源源资格检测】

LDO 上电后，器件检测输入源的供电能力。要启动 Buck 转换器，输入源必须满足：

1. LDO 使能后延迟约 200ms；
2. VBUS 低于 VBUS_OVP 阈值；
3. BUS 引脚吸收 30mA 时，VBUS 仍高于 3.8V。

若 30ms 内未通过劣质源检测，器件每 2s 重新进行一次资格检测。输入源通过全部条件后，状态寄存器 VBUS_GD 位置 1，并在 nINT 输出脉冲通知主机。

【输入源类型检测】

VBUS_GD 置位后，器件通过 DP/DM 进行输入源检测。ETA6963 按 USB Battery Charging Specification BC1.2 检测 SDP/CDP/DCP 和非标准适配器。检测完成后会更新输入限流设置及 VBUS_STAT 状态位。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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FUNCTION DESCRIPTION

ETA6963 device is a highly integrated 3A switch-mode battery charger for single cell Li-Ion and Li-polymer battery. It includes the input reverse-blocking FET (RBFET, Q1), high-side switching FET (HSFET, Q2), low-side switching FET (LSFET, Q3), and battery FET (BATFET, Q4), and bootstrap diode for the high-side gate drive

POWER UP FROM BATTERY WITHOUT INPUT SOURCE

When a battery is applied without other input, the device checks the battery voltage to turn on BATFET but not V_{LDO} to minimize the supply current, especially for NTC resistor network. All supply will be directly from battery. V_{LDO} is disabled until BOOST is enabled. Anyway, in order to optimize the discharge current from battery at light load, BATFET gate charge pump operates with lower frequency.

POWER UP WITH INPUT SOURCE

When an input source is plugged in, the device checks the input source voltage to turn on LDO and all the bias circuits. It detects and sets the input current limit before the buck converter is started. The power up sequence from input source is as listed:

1. Power up V_{LDO}
2. Poor Source Qualification
3. Input Source Type Detection is based on D+/D- to set default input current limit (I_{NDPM}) register or input source type.
4. Input Voltage Limit Threshold Setting (V_{NDPM} threshold)
5. Converter Power-up

LDO POWER UP

V_{LDO} is enabled when the below conditions are met:

1. V_{BUS} is above 3.65V
2. V_{BUS} is above $V_{BAT} + V_{SLEEP}$ (200mV) in Buck mode or V_{BUS} below $V_{BAT} + V_{SLEEP}$ in Boost mode

Right after two above conditions are valid, LDO is powered up right away

POWER SOURCE QUALIFICATION

After LDO powers up, the device checks the current capability of the input source. The input source has to meet the following requirements in order to start the buck converter.

1. 200mS delay after LDO enable
2. V_{BUS} below the V_{BUS_OVP} threshold
3. V_{BUS} above 3.8V when pulling down 30mA at BUS pin

If the device fails the poor source detection in 30ms, it repeats poor source qualification every 2 seconds. Once the input source passes all the conditions above, the status register bit $VBUS_GD$ is set high and the $nINT$ pin is pulsed to signal to the host.

INPUT SOURCE TYPE DETECTION

After the $VBUS_GD$ bit is set, the device will run input source detection through the DP/DM lines. The ETA6963 follows the USB Battery Charging Specification (BC1.2) to detect input source (SDP/CDP/DCP) and non-standard adapter through USB DP/DM lines. And after input type detection is completed, Input current limit register is changed

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【DP/DM 检测设置输入限流】

主机可根据需要改写 IINDPM 寄存器以调整输入电流限制，充电器输入电流始终受 IINDPM 限制。

ETA6963 在 VBUS 插入时通过 DP/DM 检测输入源并设置输入限流。检测包括标准 USB BC1.2 与非标准适配器。首先执行 USB BC1.2 检测，可识别标准下行端口（SDP）、充电下行端口（CDP）和专用充电端口（DCP）。当数据接触检测 DCD 定时结束后，再执行非标准适配器检测，根据 DP/DM 上特定分压识别 Apple、Samsung 等厂商适配器。

【表 1 非标准适配器检测】

类型	DP 阈值	DM 阈值	输入限流
Divider1	VDP 在 V2P7_VTH 范围	VDM 在 V2P0_VTH 范围	2.1A
Divider2	VDP 在 V1P2_VTH 范围	VDM 在 V1P2_VTH 范围	2.0A
Divider3	VDP 在 V2P0_VTH 范围	VDM 在 V2P7_VTH 范围	1.0A
Divider4	VDP 在 V2P7_VTH 范围	VDM 在 V2P7_VTH 范围	2.4A

【表 2 D+/D- 检测得到的输入限流】

USB SDP：0.5A
USB CDP：1.5A
USB DCP：2.4A
Divider1：2.1A
Divider2：2.0A
Divider3：1.0A
Divider4：2.4A
未知适配器：0.5A

【Buck 上电】

设置输入电流后，器件启动 Buck 转换器并允许 HSFET/LSFET 开关。ETA6963 提供软启动和 VSYS 短路保护，以避免过冲电流。软启动时间结束后，BATFET 开始导通，允许进入充电过程。

系统电源轨启动时也采用软启动。当系统电压低于 2.2V 时，输入限流取 200mA 与 IINDPM 设置值中的较小值；系统电压高于 2.2V 后，输入限流恢复为 IINDPM 设置值。作为电池充电器，器件采用高效率 1.5MHz 降压开关稳压架构。

原页图示参考

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to set current limit, VBUS_STAT bits are updated.

The host can over-write IINDPM register to change the input current limit if needed. The charger input current is always limited by the IINDPM register.

DP/DM DETECTION SETS INPUT CURRENT LIMIT

The ETA6963 contains a DP/DM based input source detection to set the input current limit at V_{BUS} plug-in. The DP/DM detection includes standard USB BC1.2 and non-standard adapter. When input source is plugged in, the device starts standard USB BC1.2 detections. The USB BC1.2 is capable to identify Standard Downstream Port (SDP), Charging Downstream Port (CDP) and Dedicated Charging Port (DCP). When the Data Contact Detection (DCD) timer expires, the non-standard adapter detection is applied to set the input current limit. The non-standard detection is used to distinguish vendor specific adapters (Apple and Samsung) based on their unique dividers on the DP/DM pins. Base on DP/DM configuration, the input current limit will be set following the below table:

Table 1: Non-standard Adapter Detection

NON-STANDARD ADAPTER	DP THRESHOLD	DM THRESHOLD	INPUT CURRENT LIMIT (A)
Divider1	V _{DP} within V2P7_VTH	V _{DM} within V2P0_VTH	2.1
Divider2	V _{DP} within V1P2_VTH	V _{DM} within V1P2_VTH	2
Divider3	V _{DP} within V2P0_VTH	V _{DM} within V2P7_VTH	1
Divider4	V _{DP} within V2P7_VTH	V _{DM} within V2P7_VTH	2.4

Table 2: Input Current Limit Setting from D+/D- Detection

D+/D- DETECTION	INPUT CURRENT LIMIT (A)
USB SDP	0.5
USB CDP	1.5
USB DCP	2.4
Divider1	2.1
Divider2	2
Divider3	1
Divider4	2.4
Unknown Adapter	0.5

BUCK POWER UP

After the input current is set, the device starts Buck converter and allow HSFET and LSFET switching. The ETA6963 provide soft-start time, V_{SYS} short protection to avoid overshoot current. When switching is over the soft-start time, the BATFET starts turning on then allows charging progress.

The device provides soft-start when system rail is ramped up. When the system rail is below 2.2 V, the input current limit is set to the lower of 200 mA or IINDPM register setting. After the system rises above 2.2 V, the device limits input current to the value of IINDPM register. As a battery charger, the device deploys a highly efficient 1.5 MHz

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【Buck 工作（续）】

固定频率振荡器在输入电压、电池电压、充电电流和温度变化时仍能较严格地控制开关频率，从而简化输出滤波器设计。为提高轻载效率，当电池电压低于最小系统电压设置或充电被禁止时，器件在轻载转入 PFM 控制。PFM 工作期间，开关占空比由 SYS 与 BUS 的电压比决定。PFM_DIS 位可在 Buck 或 Boost 配置下禁止 PFM。

【Boost 上电】

ETA6963 提供最高约 1.2A 输出电流的 Boost 转换器。满足以下条件时可使能 Boost：

1. VBAT 高于退出 Boost 模式的电池电压阈值；
2. VBUS < VBAT + VSLEEP（约 200mV）；
3. OTG_CONFIG=1；
4. TS 引脚位于允许温度范围：VBHOT < VTS < VBCOLD。

Boost 模式下，VBUS_STAT=111，BUS 默认输出约 5.15V，开关频率 1.5MHz。输出电流可通过 I2C 的 BOOST_LIM 位选择，最高约 1.2A。只要 VBAT 高于 VOTG_BAT 阈值，Boost 输出就会保持。启用 OTG 时，器件先以 PFM 启动，随后切换到 PWM 以降低过冲；PFM_DIS 同样可禁止 PFM。

【主机模式与默认模式】

ETA6963 虽然是主机控制型充电器，但也可以在无主机管理时以默认模式自主工作。默认模式适合完全无主机或主机处于睡眠状态的场景。

默认模式下 WATCHDOG_FAULT=1；主机模式下 WATCHDOG_FAULT=0。

上电复位后，器件以看门狗已超时/默认模式启动，全部寄存器采用默认值。在默认模式下，器件使用默认 10 小时快充安全计时器持续充电。10 小时结束后停止充电，但 Buck 转换器继续为系统负载供电。

向 WD_RST 写 1 可从默认模式切换到主机模式，之后主机可编程所有器件参数。若要保持主机模式，主机必须在看门狗到期前向 WD_RST 写 1 重置看门狗，或者将 WATCHDOG 位设置为 00 禁用看门狗。

看门狗超时（WATCHDOG_FAULT=1）后，器件返回默认模式并将寄存器恢复默认值，但以下位/字段例外：IINDPDM、VINDPDM、BATFET_DLY、BATFET_DIS、EN_ICHG_MON、PFM_DIS、SYS_Min、Min_VBAT_SEL、Q1_FULLON、OVP、BOOSTV、VDPM_BAT_TRACK、VINDPDM_INT_MASK。

原页图示参考

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step-down switching regulator. The fixed frequency oscillator keeps tight control of the switching frequency under all conditions of input voltage, battery voltage, charge current and temperature, simplifying output filter design.

In order to improve light load efficiency, the device switches to PFM control at light load when battery is below minimum system voltage setting or charging is disabled. During the PFM operation, the switching duty cycle is set by the voltage ratio of SYS and BUS. The PFM_DIS bit can be used to prevent PFM operation in either buck or boost configuration.

BOOST POWER UP

ETA6963 provide boost converter up to 1.2A output current. The boost is enabled when the below conditions are valid:

1. V_{BAT} above the battery voltage exiting boost mode threshold
2. V_{BUS} is less than $V_{BAT} + V_{SLEEP}$ (200mV)
3. OTG_CONFIG bit=1
4. TS pin is within acceptable range ($V_{BHOT} < V_{TS} < V_{BCOLD}$)

During boost mode, the status register VBUS_STAT bits is set to 111, the BUS output voltage is 5.15 V by default and 1.5MHz Frequency. The output current can reach up to 1.2 A, selected through I2C (BOOST_LIM bit). The boost output is maintained when V_{BAT} is above V_{OTG_BAT} threshold. When OTG is enabled, the device starts up with PFM and later transits to PWM to minimize the overshoot. The PFM_DIS bit can be used to prevent PFM operation in either buck or boost configuration.

HOST MODE AND DEFAULT MODE

The ETA6963 is a host controlled charger, but it can operate in default mode without host management. In default mode, the device can be used as an autonomous charger with no host or while host is in sleep mode. When the charger is in default mode, WATCHDOG_FAULT bit is logic high. When the charger is in host mode, WATCHDOG_FAULT bit is logic low.

After power-on-reset, the device starts in default mode with watchdog timer expired or default mode. All the registers are in the default settings.

In default mode, the device keeps charging the battery with default 10 hours fast charging safety timer. At the end of the 10 hours, the charging is stopped and the buck converter continues to operate to supply system load.

Writing a '1' to the WD_RST bit transitions the charger from default mode to host mode. All the device parameters can be programmed by the host. To keep the device in host mode, the host has to reset the watchdog timer by writing '1' to WD_RST bit before the watchdog timer expires (WATCHDOG_FAULT bit is set) or disable watchdog timer by setting WATCHDOG bits = 00.

When the watchdog timer expires (WATCHDOG_FAULT bit = 1), the device returns to default mode and all registers are reset to default values except IINDPM, VINDPM, BATFET_DLY, BATFET_DIS, EN_ICHG_MON, PFM_DIS, SYS_Min, Min_VBAT_SEL, Q1_FULLON, OVP, BOOSTV, VDPM_BAT_TRACK, and VINDPM_INT_MASK bits.

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【看门狗流程图】

流程图说明默认模式与主机模式之间的切换：POR 后进入默认模式并复位看门狗/寄存器；I2C 写入且 WD_RST=1 时进入主机模式并启动看门狗；若看门狗超时则返回默认模式。详细状态跳转见原页图 1。

【自主电池充电器】

器件可为单节锂离子电池提供最高约 3A 充电电流，适合大容量平板电池。约 27mohm 的 BATFET 有助于提高充电效率并降低放电时压降。

当充电使能（CHG_CONFIG=1 且 nCE=低）时，器件无需主机参与即可自主完成完整充电周期。主机也可以随时通过 I2C 写相应寄存器来控制充电并优化参数。

满足以下条件时开始新的充电周期：

- 转换器已启动；
- 充电使能：CHG_CONFIG=1、ICFG 寄存器不为 0mA、CE 为低；
- TS 无热敏电阻故障；
- 无安全计时器故障；
- BATFET 未被强制关闭，即 BATFET_DIS=0。

【表 3 默认充电参数】

充电电压：4.2V

充电电流：2.048A

预充电流：180mA

终止电流：180mA

温度策略：JEITA

安全计时器：10 小时

当充电电流低于终止阈值、充电电压高于再充电阈值，并且器件未处于 DPM 或热调节状态时，充电器自动终止充电周期。

满充电池电压下降到再充电阈值以下时（由 VRECHG 位选择），器件自动开始新的充电周期。充电完成后，也可通过翻转 nCE 或 CHG_CONFIG 位重新启动充电周期。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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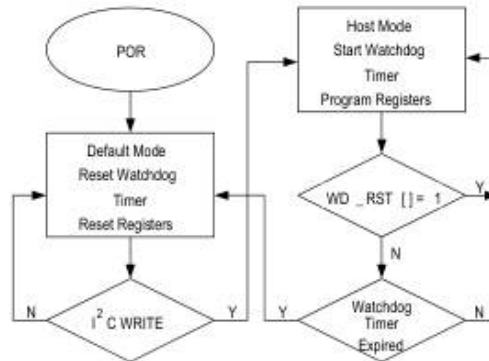


Figure 1: Watchdog Timer Flow Chart

AUTONOMOUS BATTERY CHARGER

The device charges 1cell Li-Ion battery with up to 3A charge current for high capacity tablet battery. The 27mΩ BATFET improves charging efficiency and minimize the voltage drop during discharging.

With battery charging is enabled (CHG_CONFIG = 1 and nCE pin is LOW), the device autonomously completes a charging cycle without host involvement. The host can always control the charging operations and optimize the charging parameters by writing to the corresponding registers through I²C.

A new charge cycle starts when the following conditions are valid:

- Converter starts
- Battery charging is enabled (CHG_CONFIG bit = 1 and ICHG register is not 0 mA and CE is low)
- No thermistor fault on TS
- No safety timer fault
- BATFET is not forced to turn off (BATFET_DIS bit = 0)

Table 3: Charging Parameter Default Setting

Default Mode	ETA6963
Charging Voltage	4.2V
Charging Current	2.048A
Pre-charge Current	180mA
Termination Current	180mA
Temperature Profile	JEITA
Safety Timer	10 hours

The charger device automatically terminates the charging cycle when the charging current is below termination threshold, charge voltage is above recharge threshold, and device not is in DPM mode or thermal regulation.

When a full battery voltage is discharged below recharge threshold (selectable through VRECHG bit), the device automatically starts a new charging cycle. After the charge is done, toggle nCE pin or CHG_CONFIG bit can initiate a new charging cycle.

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【充电状态】

STAT 输出用于指示充电状态：充电中为 LOW；充电完成或禁止充电为 HIGH；充电故障时闪烁。将 EN_ICHG_MON 设置为 11 可关闭 STAT 输出。

状态寄存器 CHRG_STAT 对应充电阶段：

00：禁止充电；

01：预充；

10：快充（恒流）及恒压阶段；

11：充电完成。

完成一次充电周期后，会输出 INT 脉冲通知主机。

【电池充电曲线】

器件按五个阶段为电池充电：电池短路涓流、预处理、恒流、恒压和可选 Top-Off 涓流充电。每个充电周期开始时，器件检测电池电压并据此调节电流和电压。

若充电过程中器件处于 DPM 调节或热调节状态，实际充电电流会低于编程值。此时暂时禁止充电终止判断，同时充电安全计时器按一半时钟速率计数。

图 2 给出了 SYS 电压、BAT 电压和电池充电电流在涓流、预充、快充、恒压、Top-Off 和总计时器到期阶段的变化。关键参数包括 VREG、ICHG_REG、V_{SYS_MIN}、V_{BAT_LOWV}=3.12V、V_{SHORT}≈2.25V、I_{PRECHG}、I_{TERM}、I_{SHORT}≈145mA。

【充电终止配置】

当电池电压达到 VREG[4:0] 设置的稳压值且 EN_TERM=1 时，允许进入终止判断。此时充电电流降至 I_{TERM} 以下后，在 T_{TOPOFF} 后结束充电。充电周期完成后 BATFET 关闭，转换器继续为系统供电；BATFET 仍可在需要时重新导通进入补充模式。

终止发生时 CHRG_STAT=11，并向主机产生 INT 脉冲。若器件处于输入电流调节、输入电压调节或热调节状态，则终止判断暂时禁止。可在终止前将 EN_TERM 写 0 完全禁用终止功能。

在较低终止电流下，比较器失调可能导致实际终止电流高于目标值。为补偿失调，可在检测到终止后启用可编程 Top-Off 计时器。Top-Off 计时器受安全计时器约束：安全计时器暂停时，Top-Off 也暂停；安全计时器按 2 倍时间计数时，Top-Off 同样按对应比例延长。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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The STAT output indicates the charging status: charging (LOW), charging complete or charge disable (HIGH) or charging fault (Blinking). The STAT output can be disabled by setting EN_ICHG_MON bits = 11. In addition, the status register (CHRG_STAT) indicates the different charging phases:

- 00-charging disable
- 01-precharge
- 10-fast charge (constant current) and constant voltage mode
- 11-charging done

Once a charging cycle is completed, an INT pulse is asserted to notify the host.

BATTERY CHARGER PROFILE

The device charges the battery in five phases: battery short, pre-conditioning, constant current, constant voltage and top-off trickle charging (optional). At the beginning of a charging cycle, the device checks the battery voltage and regulates current and voltage accordingly.

If the charger device is in DPM regulation or thermal regulation during charging, the actual charging current will be less than the programmed value. In this case, termination is temporarily disabled and the charging safety timer is counted at half the clock rate.

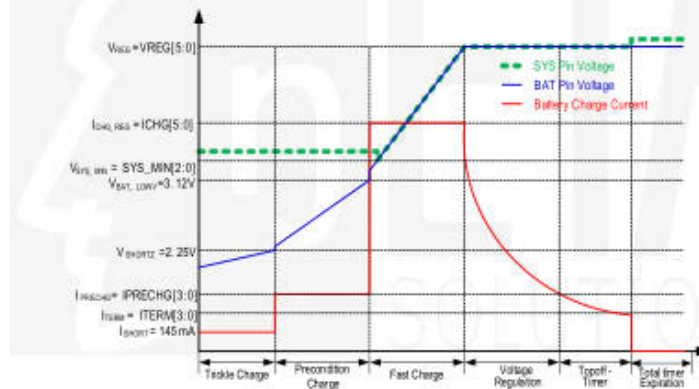


Figure 2: Battery Charger Profile

CHARGE TERMINATION CONFIGURATION

When battery voltage reach regulation level set by VREG[4:0], the device allow charger enter termination if EN_TERM[] = 1. In this state, when charge current falls below I_TERM level, charger will be terminated after T_TOPOFF. After the charging cycle is completed, the BATFET turns off. The converter keeps running to power the system, and BATFET can turn on again to engage Supplement Mode.

When termination occurs, the status register CHRG_STAT is set to 11, and an INT pulse is asserted to the host. Termination is temporarily disabled when the charger device is in input current, voltage or thermal regulation. Termination can be disabled by writing 0 to EN_TERM bit prior to charge termination.

At low termination currents, due to the comparator offset, the actual termination current may be higher than the termination target. In order to compensate for comparator offset, a programmable top-off timer can be applied after termination is detected. The termination timer will follow safety timer constraints, such that if safety timer is suspended, so will the termination timer. Similarly, if safety timer is doubled, so will the termination timer.

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【Top-Off 定时器】

TOPOFF_ACTIVE 位用于报告 Top-Off 定时器是否正在运行。主机可结合 CHRG_STAT 和 TOPOFF_ACTIVE 判断充电终止状态。

以下任一条件会复位 Top-Off 定时器：

- 从禁止充电变为使能充电；
- 终止状态由低变高；
- REG_RST 位置位。

Top-Off 时间设置在充电器检测到终止时锁存。若在终止之后再修改 Top-Off 值，除非启动新的再充电周期，否则不会生效。进入 Top-Off 阶段以及 Top-Off 计时结束时都会向主机输出 INT 脉冲。

【充电安全计时器】

器件内置安全计时器，防止异常电池导致充电周期无限延长。当电池低于 VBATLOW 时，安全计时器为 2 小时。快充安全时间可通过 I2C 的 CHG_TIMER 位编程。安全计时到期后，故障寄存器 CHRG_FAULT=11，并产生 INT 脉冲。通过 EN_TIMER 位可关闭安全计时功能。

在输入电压调节、输入电流调节、JEITA 冷温区或热调节期间，因为实际充电电流通常低于寄存器设定值，安全计时器以一半时钟速率计数。例如安全时间设为 5 小时且整个周期始终处于输入电流调节状态，则实际约 10 小时才超时。将 TMR2X_EN 写 0 可关闭该“半速计时”功能。

发生故障时计时器暂停；故障解除后继续计时。若用户停止当前充电并重新开始，例如翻转 nCE 或 CHRG_CONFIG，计时器会复位。

【最小 SYS 电压】

由于 SYS 电源轨直接为系统供电，ETA6963 提供最小 SYS 电压设置，确保系统始终获得足够电压。最小系统电压由 SYS_MIN[2:0] 设置，默认约 3.5V。即使电池完全耗尽，系统电压仍被调节在最小系统电压以上。

当电池电压低于最小系统电压设置时，BATFET 在线性/LDO 模式工作，SYS 通常比最小系统电压高约 180mV。电池电压升至最小系统电压以上后，BATFET 完全导通，SYS 与 BAT 的差值主要为 BATFET 的 VDS。

当充电被禁止且电池电压高于最小系统电压，或充电已经终止时，SYS 通常调节为约 VBAT+70mV。当处于最小系统电压调节时，状态位 VSYS_STAT 置高。

图 3 给出了系统电压随电池电压的关系，并对比最小系统电压、充电禁止和充电使能状态。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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TOPOFF_ACTIVE bit reports whether the top off timer is active or not. The host can read CHRG_STAT and TOPOFF_ACTIVE to find out the termination status.

Top off timer gets reset at one of the following conditions:

- Charge disable to enable
- Termination status low to high
- REG_RST register bit is set

The top-off timer settings are read in once termination is detected by the charger. Programming a top-off timer value after termination will have no effect unless a recharge cycle is initiated. An INT pulse is asserted to the host when entering top-off timer segment as well as when top-off timer expires.

CHARGE SAFETY TIMER

The device has built-in safety timer to prevent extended charging cycle due to abnormal battery conditions. The safety timer is 2 hours when the battery is below V_{BATLOW} threshold. The user can program fast charge safety timer through I²C (CHG_TIMER bits). When safety timer expires, the fault register CHRG_FAULT bits are set to 11 and an INT pulse is asserted to the host. The safety timer feature can be disabled through I²C by setting EN_TIMER bit.

During input voltage, current, JEITA cool or thermal regulation, the safety timer counts at half clock rate as the actual charge current is likely to be below the register setting. For example, if the charger is in input current regulation (IDPM_STAT = 1) throughout the whole charging cycle, and the safety time is set to 5 hours, the safety timer will expire in 10 hours. This half clock rate feature can be disabled by writing 0 to TMR2X_EN bit.

During the fault, timer is suspended. Once the fault goes away, fault resumes. If user stops the current charging cycle, and start again, timer gets reset (toggle nCE pin or CHRG_CONFIG bit).

MINIMUM SYS VOLTAGE

Because the rail that provide power to system is SYS voltage, so to make sure the system always has enough voltage, ETA6963 provide minimum SYS voltage selection. The minimum system voltage is set by SYS_MIN[2:0] bits. Even with a fully depleted battery, the system is regulated above the minimum system voltage (default 3.5 V).

When the battery is below minimum system voltage setting, the BATFET operates in linear mode (LDO mode), and the system is typically 180 mV above the minimum system voltage setting. As the battery voltage rises above the minimum system voltage, BATFET is fully on and the voltage difference between the system and battery is the V_{DS} of BATFET.

When the battery charging is disabled and above minimum system voltage setting or charging is terminated, the system is always regulated at typically 70mV above battery voltage. The status register VSYS_STAT bit goes high when the system is in minimum system voltage regulation.

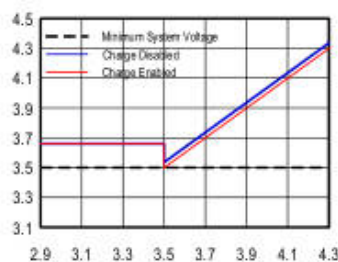


Figure 3: System Voltage vs Battery Voltage

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【动态功率管理 DPM】

为符合 USB 最大电流限制并避免适配器过载，器件提供动态功率管理（DPM），持续监测输入电流和输入电压。当输入源过载时，可能表现为电流超过 IINDPM 或输入电压低于 VINDPM。器件会自动降低充电电流，直到输入电流低于限值且输入电压恢复到限压值以上。

如果充电电流已经降为 0，而输入源仍然过载，则系统电压开始下降。一旦 SYS 低于 BAT，器件自动进入补充模式，BATFET 打开，由输入源和电池共同为系统供电。

DPM 工作时，状态位 VDPM_STAT（电压 DPM）或 IDPM_STAT（电流 DPM）置高。

【补充模式】

当 SYS 比 BAT 低约 10mV（VBAT>VSYSMIN 且 Q4 已完全导通，此时放电电流约 1A），或低约 20mV（VBAT<VSYSMIN 且 Q4 正处于调节状态，此时 Q4 栅压接近阈值、放电电流接近 0）时，BATFET 开启。低电流时，BATFET 栅极驱动被调节，使其 VDS 最小保持在约 30mV，以避免在补充模式进入/退出边界发生振荡。随着放电电流增大，BATFET 栅极电压继续升高以降低 RDSON，直至完全导通。完全导通后，BATFET 的 VDS 大体随放电电流线性增加。图 4 给出了进入补充模式时 BATFET 的 V-I 特性。电池电压低于电池耗尽阈值时，BATFET 关闭并退出补充模式。

【运输模式与 nQON】

【BATFET 关闭模式（运输模式）】

系统空闲、运输或存储期间，为延长电池寿命并降低关机功耗，可关闭 BATFET，使 SYS 变为 0，从而减小电池漏电。主机设置 BATFET_DIS 后，可立即关闭 BATFET，或者按 BATFET_DLY 配置延迟 tSM_DLY 后关闭。

【BATFET 使能（退出运输模式）】

BATFET 关闭且 BATFET_DIS 已置位时，以下事件之一可重新使能 BATFET、恢复系统供电：

1. 插入适配器；
2. 清除 BATFET_DIS；
3. 设置 REG_RST，复位所有寄存器，包括将 BATFET_DIS 恢复为默认 0；
4. nQON 由高到低并保持超过 tSHIPMODE 去毛刺时间（典型约 1.15s）。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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DYNAMIC POWER MANAGEMENT

To meet maximum current limit in USB spec and avoid over loading the adapter, the device features Dynamic Power management (DPM), which continuously monitors the input current and input voltage. When input source is overloaded, either the current exceeds the input current limit (I_{NDPM}) or the voltage falls below the input voltage limit (V_{NDPM}). The device then reduces the charge current until the input current falls below the input current limit and the input voltage rises above the input voltage limit.

When the charge current is reduced to zero, but the input source is still overloaded, the system voltage starts to drop. Once the system voltage falls below the battery voltage, the device automatically enters the supplement mode where the BATFET turns on and battery starts discharging so that the system is supported from both the input source and battery.

During DPM mode, the status register bits VDPM_STAT (V_{NDPM}) or IDPM_STAT (I_{NDPM}) goes high.

SUPPLEMENT MODE

When the system voltage falls 10 mV ($V_{BAT} > V_{SYSMIN}$ when Q4 is full on. In this condition, discharge current is almost 1A.) or 20mV ($V_{BAT} < V_{SYSMIN}$ when Q4 is in regulation. In this condition, gate of Q4 is regulated at about only threshold voltage of the FET, then discharge current is almost zero.) below the battery voltage, the BATFET turns on and the BATFET gate is regulated the gate drive of BATFET so that the minimum BATFET V_{DS} stays at 30 mV when the current is low. This prevents oscillation from entering and exiting the supplement mode.

As the discharge current increases, the BATFET gate is regulated with a higher voltage to reduce $R_{DS(on)}$ until the BATFET is in full conduction. At this point onwards, the BATFET V_{DS} linearly increases with discharge current. Following figure shows the V-I curve of the BATFET gate regulation operation. BATFET turns off to exit supplement mode when the battery is below battery depletion threshold.

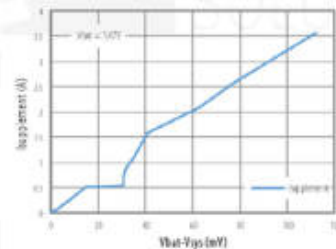


Figure 4: BATFET Current at Entering Supplement Mode

SHIPPING MODE AND nQON PIN

BATFET DISALBE MODE (SHIPPING MODE)

To extend battery life and minimize power when system is powered off during system idle, shipping, or storage, the device can turn off BATFET so that the system voltage is zero to minimize the battery leakage current. When the host set BATFET_DIS bit, the charger can turn off BATFET immediately or delay by t_{SM_DLY} as configured by BATFET_DLY bit.

BATFET ENABLE (EXIT SHIPPING MODE)

When the BATFET is disabled (in shipping mode) and indicated by setting BATFET_DIS, one of the following events can enable BATFET to restore system power:

1. Plug in adapter
2. Clear BATFET_DIS bit
3. Set REG_RST bit to reset all registers including BATFET_DIS bit to default (0)

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【BATFET 整机复位】

无输入源时，BATFET 相当于电池与系统之间的负载开关。将 BATFET 从导通切换为关闭，可强制连接到 SYS 的系统发生一次上电复位。nQON 支持按钮接口，无需主机即可通过改变 BATFET 状态来复位系统电源。

当没有接入输入源、BATFET 已使能 (BATFET_DIS=0)，且 nQON 保持低电平达到 tQON_RST 时，BATFET 会关闭 tBATFET_RST，再重新使能，从而实现系统电源复位。将 BATFET_RST_EN 设置为 0 可禁用该功能。

【nQON 引脚操作】

内部从 nQON 到 VDD (VBAT 与 VLDO 中较大者) 有约 500kohm 上拉，因此默认 nQON 为高。nQON 逻辑低/高阈值约为 0.68V/0.88V。该引脚具有两种 BATFET 控制功能：

1. BATFET 使能：nQON 从高到低并保持超过 tSHIPMODE，可打开 BATFET 并退出运输模式。
2. BATFET 复位：未插适配器且 BATFET_DIS=0 时，nQON 保持低电平至少 tQON_RST (文中典型约 12s)，BATFET 关闭 tBATFET_RST 后重新使能，使 SYS 发生一次上电复位。可通过 BATFET_RST_EN=0 禁用。

图 5 给出了 nQON 按键、tSHIPMODE、tQON_RST 和 tBATFET_RST 的时序。

【充电模式下热敏电阻温度判定】

充电器提供一个热敏电阻输入用于监测电池温度。典型电路从 LDO 通过 RTS1 到 TS，再通过 RTS2/NTC 网络到地，具体见原页。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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- A logic high to low transition on nQON pin with t_{SHIPMODE} deglitch time (1.15s typical) to enable BATFET to exit shipping mode

BATFET FULL SYSTEM RESET

The BATFET functions as a load switch between battery and system when input source is not plugged-in. By changing the state of BATFET from on to off, systems connected to SYS can be effectively forced to have a power-on-reset. The nQON pin supports push-button interface to reset system power without host by changing the state of BATFET.

When the QON pin is driven to logic low for $t_{\text{QON_RST}}$ while input source is not plugged in and BATFET is enabled (BATFET_DIS = 0), the BATFET is turned off for $t_{\text{BATFET_RST}}$ and then it is re-enabled to reset system power. This function can be disabled by setting BATFET_RST_EN bit to 0.

nQON PIN OPERATION

Internally, there is a 500K pull up resistor from nQON pin to V_{DD} (maximum of V_{BAT} and V_{LDO}). So default, nQON will be high. Then nQON logic low and high voltage threshold are 0.68V and 0.88V respectively. The nQON pin incorporates two functions to control BATFET.

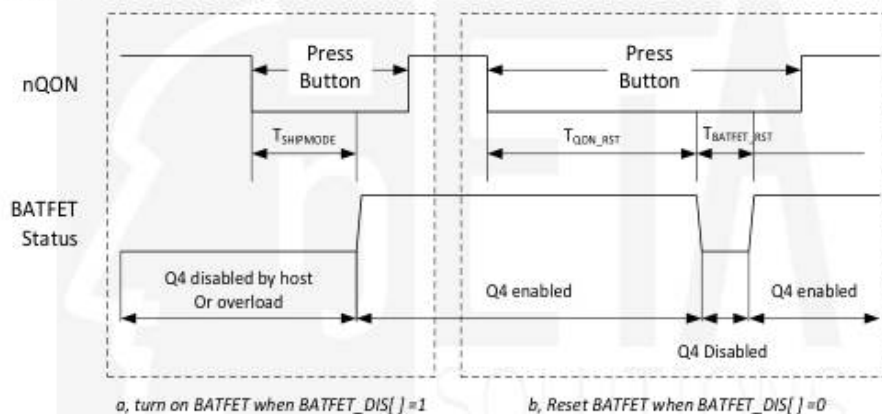
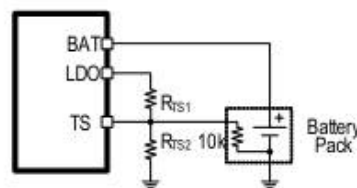


Figure 5: nQON Timing

- BATFET Enable: A nQON logic transition from high to low with longer than t_{SHIPMODE} deglitch turns on BATFET and exit shipping mode
- BATFET Reset: When nQON is driven to logic low by at least $t_{\text{QON_RST}}$ (12s typical) while adapter is not plugged in (and BATFET_DIS = 0), the BATFET is turned off for $t_{\text{BATFET_RST}}$ (10s typical). The BATFET is re-enabled after $t_{\text{BATFET_RST}}$ duration. This function allows systems connected to SYS to have power-on-reset. This function can be disabled by setting BATFET_RST_EN bit to 0.

THERMISTOR QUALIFICATION DURING CHARGING MODE

The charger device provides a single thermistor input for battery temperature monitor.



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【JEITA 温度策略】

为提高锂离子电池充电安全性，JEITA 于 2007 年 4 月 20 日发布相关指南，强调在特定低温和高温范围内应避免使用过高充电电流和过高充电电压。

要启动充电周期，TS 引脚电压必须位于 VT1~VT5 阈值之间。若 TS 电压超出 T1~T5 范围，控制器暂停充电，直到电池温度回到允许范围。

在低温 T1~T2 区间，JEITA 建议将充电电流降低到快充电流的一半或更低；在高温 T3~T5 区间，建议将充电电压降低到 4.1V 以下。

ETA6963 允许比 JEITA 要求更灵活的电压/电流设置：高温 T3~T5 区间的充电电压可设置为 VREG 或 4.1V（JEITA_VSET）；低温 T1~T2 区间的充电电流可进一步降低至快充电流的 20%（JEITA_ISET）。

图 7 给出 JEITA 充电电流曲线，图 8 给出 JEITA 充电电压曲线。

【热敏电阻偏置网络】

原文给出了 RTS1、RTS2 与 RTHCOLD、RTHHOT、VT1、VT5、VLDO 之间的计算公式。公式和变量符号保持原文，请参照原页核对。

【电池放电/Boost 模式温度判定】

为在 Boost 模式下保护电池，器件监测电池温度是否位于 VBCOLD~VBHOT 范围。温度超出范围时，Boost 暂停，VBUS_STAT 置 000，并报告 NTC_FAULT。温度恢复后 Boost 自动恢复，同时清除 NTC_FAULT。图 9 给出 Boost 模式 TS 热敏检测阈值。

【STAT 充电状态指示】

器件通过开漏 STAT 引脚指示充电状态，可直接驱动 LED。设置 STAT_CONFIG[1:0]=11 可关闭 STAT 功能；当 STAT_CONFIG[1:0]=00 时，按下一页表格工作。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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To improve the safety of charging Li-ion batteries, JEITA guideline was released on April 20, 2007. The guideline emphasized the importance of avoiding a high charge current and high charge voltage at certain low and high temperature ranges.

To initiate a charge cycle, the voltage on TS pin must be within the V_{T1} to V_{T5} thresholds. If TS voltage exceeds the T1-T5 range, the controller suspends charging and waits until the battery temperature is within the T1 to T5 range. At cool temperature (T1-T2), JEITA recommends the charge current to be reduced to half of the charge current or lower. At warm temperature (T3-T5), JEITA recommends charge voltage less than 4.1V.

The charger provides flexible voltage/current settings beyond the JEITA requirement. The voltage setting at warm temperature (T3-T5) can be V_{REG} or 4.1V (configured by JEITA_VSET). The current setting at cool temperature (T1-T2) can be further reduced to 20% of fast charge current (JEITA_ISET).

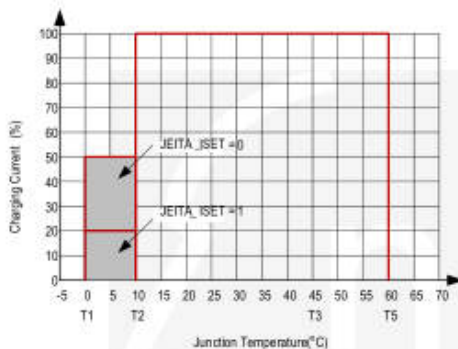


Figure 7: JEITA Profile: Charging Current

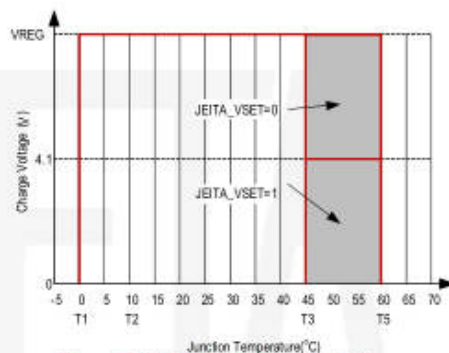


Figure 8: JEITA Profile: Charging Voltage

The resistor bias network has been updated as below.

$$R_{TS1} = \frac{\left(\frac{V_{LDO}}{V_{T1}} - 1\right)}{\frac{1}{R_{TS2}} + \frac{1}{R_{THCOLD}}}$$

$$R_{TS2} = \frac{V_{LDO} \times R_{THCOLD} \times R_{THHOT} \times \left(\frac{1}{V_{T1}} - \frac{1}{V_{T5}}\right)}{R_{THHOT} \times \left(\frac{V_{LDO}}{V_{T5}} - 1\right) - R_{THCOLD} \times \left(\frac{V_{LDO}}{V_{T1}} - 1\right)}$$

THERMISTOR QUALIFICATION DURING BATTERY DISCHARGE MODE

For battery protection during boost mode, the device monitors the battery temperature to be within the V_{BCOLD} to V_{BHOT} thresholds. When temperature is outside of the temperature thresholds, the boost mode is suspended. In addition, V_{BUS_STAT} bits are set to 000 and NTC_FAULT is reported. Once temperature returns within thresholds, the boost mode is recovered and NTC_FAULT is cleared.

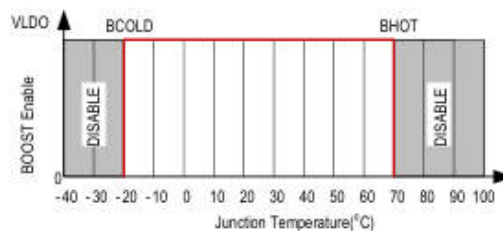


Figure 9: TS Pin Thermistor Sense Threshold in Boost

CHARGING STATUS INDICATION ON STAT PIN

The device indicates charging state on the open drain STAT pin. The STAT pin can drive LED. The STAT pin function can be disabled by setting the $STAT_CONFIG[1:0] = 11$. When $STAT_CONFIG[1:0] = 00$, STAT indication operates

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【表 4 STAT 指示】

充电中（包括再充电）：LOW。

充电完成：HIGH。

睡眠模式/禁止充电：HIGH。

充电暂停（输入过压、TS 故障、计时器故障或系统过压）：1Hz 闪烁，0.5s LOW / 0.5s 高阻。

Boost 因 TS 故障暂停：同样按故障指示处理。

【nINT 主机中断】

ETA6963 提供 nINT 脉冲，使主机无需持续轮询充电器。以下事件会产生约 256us 低有效 INT 脉冲：

- 通过 DP/DM 完成 USB/适配器源检测；
- BUS 通过劣质源检测；
- 输入移除；
- 充电完成；
- REG09 中任何故障事件；
- 检测到 VINDPM 事件（VINDPM_INT_MASK=1 时屏蔽）；
- 检测到 IINDPM 事件（IINDPM_INT_MASK=1 时屏蔽）。

发生故障时，器件发送 nINT 并把故障状态保存在 REG09，直到主机读取故障寄存器。在主机读取 REG09 且原故障未全部清除前，新故障不会再次触发 INT。要读取当前故障状态，主机应连续读取 REG09 两次：第一次返回此前锁存状态，第二次返回当前状态。

【充电模式输入过压保护 ACOV】

若 VBUS 超过由 OVP[1:0] 编程的 VBUS_OV 阈值，器件立即停止开关，CHRG_FAULT=01，并向主机产生 nINT。输入电压回落到 OVP 阈值以下后，器件自动恢复正常工作。

【充电模式系统过压保护 SYSOVP】

充电器在负载瞬态时钳位系统电压，避免连接到 SYS 的器件因过压损坏。SYSOVP 阈值约为系统调节电压的 114%。发生 SYSOVP 时，转换器立即停止以钳制过冲。

【Boost 模式 BUS 输出短路保护】

器件监测 Boost 输出电压及其他状态，提供短路和过压保护。Boost 内置精确恒流调节，可适应不同 OTG 负载。若检测到 VBUS 短路，Boost 关闭并最多重试 7 次。重试仍失败时，清除 OTG_CONFIG、禁用 OTG，同时设置 BOOST_FAULT 并产生 nINT。主机重新使能 Boost 可清除 BOOST_FAULT。

【Boost 模式 BUS 过压保护】

VBUS 超过调节目标并达到 VOTG_OVP 阈值后，器件停止开关、清除 OTG_CONFIG 并退出 Boost；后续故障状态见下一页。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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as shown as in below table.

Table 4 STAT Indication table

CHARGE STATE	STAT INDICATION
Charging in progress (including recharge)	LOW
Charging complete	HIGH
Sleep mode, charge disable	HIGH
Charge suspend (input overvoltage, TS fault, timer fault or system overvoltage)	Blinking at 1Hz (0.5s LOW / 0.5s HIZ)
Boost Mode suspend (due to TS fault)	

HOST INTERRUPT ON nINT PIN

ETA6963 provides a comfortable feature that host does not need to always monitor the charger operation. The nINT pulse notifies the system on the device operation. The following events will generate 256 μ s INT pulse.

- USB/Adapter Source Detection (through DP/DM pins)
- BUS Pass Poor Source Detection
- Input removed
- Charge Complete
- Any FAULT event in REG09
- VINDPM event detected. Masked by VINDPM_INT_MASK = 1.
- IINDPM event detected. Masked by IINDPM_INT_MASK = 1.

When a fault occurs, the charger device sends out nINT and keeps the fault state in REG09 until the host reads the fault register. Before the host reads REG09 and all the faults are cleared, the charger device would not send any INT upon new faults. To read the current fault status, the host has to read REG09 two times consecutively. The first read reports the pre-existing fault register status and the second read reports the current fault register status.

INPUT OVERVOLTAGE PROTECTION IN CHARGE MODE (ACOV)

If V_{BUS} exceeds the V_{BUS} over voltage threshold V_{BUS_OV} (programmable via OVP[1:0] bits) the device stops switching immediately. During input overvoltage event, the fault register CHRG_FAULT bits are set to 01. An nINT pulse is asserted to the host. The device will automatically resume normal operation once the input voltage drops back below the OVP threshold.

SYSTEM OVERVOLTAGE PROTECTION IN CHARGE MODE (SYSOVP)

The charger device clamps the system voltage during load transient so that the components connect to system would not be damaged due to high voltage. SYSOVP threshold is 14% above system regulation voltage. Upon SYSOVP, converter stops immediately to clamp the overshoot.

BUS OUTPUT SHORT PROTECTION IN BOOST MODE

The device monitors boost output voltage and other conditions to provide output short circuit and overvoltage protection. The Boost build-in accurate constant current regulation to allow OTG to adaptive to various types of load. If short circuit is detected on V_{BUS} , the Boost turns off and retry 7 times. If retries are not successful, OTG is disabled with OTG_CONFIG bit cleared. In addition, the BOOST_FAULT bit is set and an nINT pulse is generated. The BOOST_FAULT bit can be cleared by host by re-enabling boost mode.

BUS OVERVOLTAGE PROTECTION IN BOOST MODE

When the V_{BUS} rises above regulation target and exceeds the over voltage threshold V_{OTG_OVP} , the device enters overvoltage protection which stops switching, clears OTG_CONFIG bit and exits boost mode. At Boost overvoltage

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【Boost 过压故障】

Boost 过压期间，BOOST_FAULT 置高表示 Boost 工作故障，同时向主机产生 nINT 脉冲。

【Buck 模式热调节】

ETA6963 监测内部结温 TJ，防止芯片过热。在 Buck 模式下默认将 IC 温度限制在约 110 °C。当内部结温超过热调节阈值时，器件降低充电电流。热调节期间实际充电电流通常低于编程值，因此充电终止判断被禁止，安全计时器以半速运行，状态位 THERM_STAT 置高。

通过 TREG 位可将热调节阈值编程为约 90 °C。

【热关断】

器件在所有模式下都监测内部结温。IC 温度超过 TSHUT（典型 160 °C）时，BATFET 和转换器关闭。若正在 Boost 模式，则清除 OTG_CONFIG 并设置 BOOST_FAULT；若处于充电模式，则设置 CHARGE_FAULT，同时向主机产生 nINT。

温度下降到比 TSHUT 低 TSHUT_HYS（典型 30 °C）后，BATFET 和充电器按使能条件恢复，Boost 可由主机重新使能。

【电池过压保护】

电池过压限制约为电池调节电压的 107%。发生电池过压时，充电立即停止，BAT_FAULT 置高并产生 nINT。

【电池过放保护】

电池电压下降到耗尽阈值 VBAT_DPL_FALL 以下时，BATFET 关闭以防止过放。恢复需要在 VBUS 接入输入电源。接入后 BATFET 打开：当 VBAT < VSHORT 时用 ISHORT（典型约 150mA）涓流充电；当 VSHORT < VBAT < VBAT_LOWV 时，按 IPRECHG 设置的预充电电流充电。

【系统过流保护】

系统短路或严重过载使 IBAT 超过 IBATOP 时，器件锁存关闭 BATFET。可按“BATFET 使能/退出运输模式”章节的方法复位锁存并重新打开 BATFET。

【I2C 通信】

器件使用兼容 I2C 的双向两线接口，可灵活编程充电参数并即时读取状态。I2C 使用 SDA 数据线和 SCL 时钟线。主设备负责发起传输并产生时钟，被寻址器件作为从设备。

ETA6963 作为从设备，地址为 D6H，通过 REG00~REG0B 接收微控制器或 DSP 等主机的控制。接口支持标准模式最高 100kbit/s 和快速模式最高 400kbit/s。SDA/SCL 通过电流源或上拉电阻连接到正电源；总线空闲时两线均为 HIGH。SDA 和 SCL 均为开漏结构。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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duration, the fault register bit (BOOST_FAULT) is set high to indicate fault in boost operation. An nINT pulse is also asserted to the host.

THERMAL REGULATION IN BUCK MODE

The ETA6963 monitors the internal junction temperature T_J to avoid overheat the chip and limits the IC surface temperature at 110°C in buck mode. When the internal junction temperature exceeds thermal regulation limit (110°C), the device lowers down the charge current. During thermal regulation, the actual charging current is usually below the programmed battery charging current. Therefore, termination is disabled, the safety timer runs at half the clock rate, and the status register THERM_STAT bit goes high.

Thermal limit is programmable to 90°C by programming TREG bit.

THERMAL SHUTDOWN

The device monitors the internal junction temperature to provide thermal shutdown during any mode. When IC surface temperature exceeds T_{SHUT} (160°C), BATFET and Converter are disabled. OTG_CONFIG bit is cleared if in BOOST mode and BOOST_FAULT bit is set. In Charge mode, CHARGE_FAULT bit is set. An nINT pulse is asserted to the host.

When IC temperature is T_{SHUT_HYS} (30°C) below T_{SHUT} (160°C), The BATFET and charger are enabled following enable condition, Boost can be enabled by host.

BATTERY OVER-VOLTAGE PROTECTION

The battery overvoltage limit is clamped at 7% above the battery regulation voltage. When battery over voltage occurs, the charger device immediately disables charge. The fault register BAT_FAULT bit goes high and an nINT pulse is asserted to the host.

BATTERY OVER-DISCHARGE PROTECTION

When battery is discharged below the battery depletion falling threshold $V_{BAT_DPL_FALL}$, the BATFET is turned off to protect battery from over discharge. To recover from over discharge, an input source is required at V_{BUS} . When an input source is plugged in, the BATFET turns on. The battery is charged with I_{SHORT} (typically 150 mA) current when the V_{BAT} is smaller than the trickle charge voltage threshold V_{SHORT} , or pre-charge current as set in I_{PRECHG} register when the battery voltage is between V_{SHORT} and the pre-charge voltage threshold V_{BAT_LOWV} .

SYSTEM OVER-CURRENT PROTECTION

When the system is shorted or significantly overloaded ($I_{BAT} > I_{BATOP}$) so that its current exceeds the overcurrent limit, the device latches off BATFET. Section BATFET Enable (Exit Shipping Mode) can reset the latch-off condition and turn on BATFET.

I²C COMMUNICATION

The device uses I²C compatible interface for flexible charging parameter programming and instantaneous device status reporting. I²C™ is a bi-directional 2 wire serial interface developed by Philips Semiconductor (now NXP Semiconductors). Only two bus lines are required: a serial data line (SDA) and a serial clock line (SCL). Devices can be considered as masters or slaves when performing data transfers. A master is the device which initiates a data transfer on the bus and generates the clock signals to permit that transfer. At that time, any device addressed is considered a slave.

The device operates as a slave device with address D6H, receiving control inputs from the master device like micro controller or a digital signal processor through REG00-REG0B. The I2C interface supports both standard mode (up to 100kbps), and fast mode (up to 400kbps), connecting to the positive supply voltage via a current source or pull-up resistor. When the bus is free, both lines are HIGH. The SDA and SCL pins are open drain.

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【I2C 单字节写/读时序】

图 10：Single Write（单字节写）。主机依次发送 START、从设备地址（A[7:0]，A[0]=0 表示写）、ACK、寄存器地址 RA[7:0]、ACK、写数据 D[7:0]、ACK、STOP。

图 11：Single Read（单字节读）。主机先以写方向发送从设备地址和寄存器地址，再重复 START，以 A[0]=1 的读方向重新发送从设备地址，随后读取 D[7:0] 并以 NACK/STOP 结束。

器件地址由 A[7:0] 表示，其中 A[0]=0 为写寄存器地址；读取寄存器时 A[0]=1。寄存器地址为 RA[7:0]，寄存器数据为 D[7:0]。SDA/SCL 的位级时序、ACK/NACK 位置见原页图 10、图 11。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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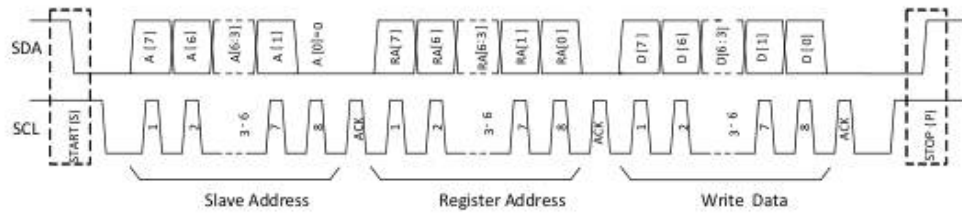


Figure 10: Single Write

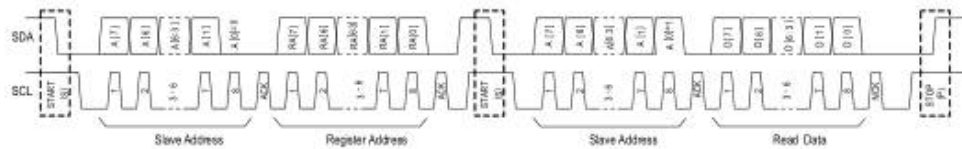


Figure 11: Single Read

Device address is understood value of A[7:0] while A[0] is "0". This is also the address to write the data to the device registers. To read data from register, sending the command to address with A[0] is "1". Register address are in RA[7:0]. Data to or from register are D[7:0].

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【寄存器映射 - REG00 / REG01（部分）】
表头：BIT | NAME | POR | TYPE | RESET BY | DESCRIPTION | NOTE。

【REG00】
Bit7 EN_HIZ, POR=0, R/W：0=关闭 HIZ, 1=使能 HIZ；默认关闭。由 REG_RST 或 Watchdog 复位。
Bit6:5 EN_ICHG_MON[1:0], POR=00, R/W：00=使能 STAT 引脚功能（默认）；01/10=保留；11=关闭 STAT 引脚功能（引脚悬空）。
Bit4:0 IINDPM[4:0], R/W, 权重分别 1600/800/400/200/100mA。输入电流限流偏置 100mA，范围 100mA（00000）～3.2A（11111），默认 2400mA（10111）。该值为最大输入限流而非典型值。输入源检测完成后器件会自动修改 IINDPM，主机随后可以覆盖写入。
【REG01（前半）】
Bit7 PFM_DIS, POR=0：0=使能 PFM（默认），1=禁用 PFM。
Bit6 WD_RST, POR=0：I2C 看门狗复位位；0=正常，1=复位看门狗；完成后自动回 0。
Bit5 OTG_CONFIG, POR=0：0=关闭 OTG（默认），1=使能 OTG。OTG_CONFIG 会覆盖 CHG_CONFIG 的充电使能功能。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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REGISTER MAP

Table 5: REG00

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	EN_HIZ	0	R/W	by REG_RST by Watchdog	0-Disable, 1-Enable	Enable HIZ Mode 0-Disable (default) 1-Enable
6	EN_ICHG_MON[1]	0	R/W	by REG_RST	00 - Enable STAT pin function (default) 01 - Reserved 10 - Reserved 11 - Disable STAT pin function (float pin)	
5	EN_ICHG_MON[0]	0				
4	IINDPM[4]	1	R/W	by REG_RST	1600 mA	Input Current Limit
3	IINDPM[3]	0	R/W	by REG_RST	800 mA	Offset: 100 mA
2	IINDPM[2]	1	R/W	by REG_RST	400 mA	Range: 100 mA (000000) – 3.2 A (11111)
1	IINDPM[1]	1	R/W	by REG_RST	200 mA	Default:2400 mA (10111), maximum input current limit, not typical.
0	IINDPM[0]	1	R/W	by REG_RST	100 mA	IINDPM bits are changed automatically after input source detection is completed Host can over-write IINDPM register bits after input source detection is completed.

Table 6: REG01

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	PFM_DIS	0	R/W	by REG_RST	0 – Enable PFM 1 – Disable PFM	Default: 0 - Enable
6	WD_RST	0	R/W	by REG_RST by Watchdog	I ² C Watchdog Timer Reset 0 –Normal; 1 – Reset	Default: Normal (0) Back to 0 after watchdog timer reset
5	OTG_CONFIG	0	R/W	by REG_RST by Watchdog	0 – OTG Disable 1 – OTG Enable	Default: OTG disable (0) Note: 1. OTG_CONFIG would over-ride Charge Enable Function in CHG_CONFIG

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【REG01（续）】

Bit4 CHG_CONFIG, POR=1 : 0=禁止充电 ; 1=使能充电。默认充电使能。注意只有 CE 引脚为低且 CHG_CONFIG=1 时才真正允许充电。

Bit3:1 SYS_Min[2:0], 系统最小电压 : 000=2.6V, 001=2.8V, 010=3.0V, 011=3.2V, 100=3.4V, 101=3.5V, 110=3.6V, 111=3.7V ; 默认 3.5V (101) 。

Bit0 Min_VBAT_SEL, POR=0 : OTG 模式最小电池电压选择。0=BAT 下降阈值 2.8V, 1=2.5V ; 默认 2.8V, 选择 0 时上升阈值约 3.0V。

【REG02】

Bit7 BOOST_LIM, POR=1 : 0=0.5A, 1=1.2A (默认) 。表中电流为最低限流规格。

Bit6 Q1_FULLON, POR=0 : 0=当编程 IINDPM<700mA 时使用较高 Q1 RDS(on) 以获得更好精度 ; 1=始终使用较低 Q1 RDS(on) 以获得更高效率。Boost 模式始终使用全导通 FET, 此位无效。

Bit5:0 ICHG[5:0], 快充电流, 位权重依次 1920、960、480、240、120、60mA ; 默认 2040mA (100010) 。范围从 0mA 到 3000mA (110010) ; ICHG=0mA 会禁止充电, 超过 3000mA 的编码钳位到 3000mA。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识, 用于与中文译文交叉核对。



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Table 6: REG01

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
4	CHG_CONFIG	1	R/W	by REG_RST by Watchdog	0 - Charge Disable 1 - Charge Enable	Default: Charge Battery (1) Note: Charge is enabled when both CE pin is pulled low AND CHG_CONFIG bit is 1
3	SYS_Min[2]	1	R/W	by REG_RST	System Minimum Voltage	000: 2.6 V 001: 2.8 V
2	SYS_Min[1]	0	R/W	by REG_RST		010: 3 V 011: 3.2 V
1	SYS_Min[0]	1	R/W	by REG_RST		100: 3.4 V 101: 3.5 V 110: 3.6 V 111: 3.7 V Default: 3.5 V (101)
0	Min_VBAT_SEL	0	R/W	by REG_RST	0 – 2.8 V BAT falling, 1 – 2.5 V BAT falling	Minimum battery voltage for OTG mode. Default falling 2.8 V (0); Rising threshold 3.0 V (0)

Table 7: REG02

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	BOOST_LIM	1	R/W	by REG_RST by Watchdog	0 = 0.5 A 1 = 1.2 A	Default: 1.2 A (1) Note: The current limit options listed are minimum current limit specs.
6	Q1_FULLON	0	R/W	by REG_RST	0 – Use higher Q1 RDSON when programmed IINDPM < 700mA (better accuracy) 1 – Use lower Q1 RDSON always (better efficiency)	In boost mode, full FET is always used and this bit has no effect
5	ICHG[5]	1	R/W	by REG_RST by Watchdog	1920 mA	Fast Charge Current Default: 2040mA (100010) Range: 0 mA (0000000) – 3000mA (110010) Note: ICHG = 0 mA disables charge. ICHG > 3000 mA (110010) clamped to register value 3000 mA (110010)
4	ICHG[4]	0	R/W	by REG_RST by Watchdog	960 mA	
3	ICHG[3]	0	R/W	by REG_RST by Watchdog	480 mA	
2	ICHG[2]	0	R/W	by REG_RST by Watchdog	240mA	
1	ICHG[1]	1	R/W	by REG_RST by Watchdog	120mA	
0	ICHG[0]	0	R/W	by REG_RST by Watchdog	60mA	

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【REG03】

Bit7:4 IPRECHG[3:0]：预充电流，位权重 480/240/120/60mA，默认 180mA（0010），步进/偏置 60mA；高于 780mA 的设置钳位到 780mA（1100）。

Bit3:0 ITERM[3:0]：终止电流，位权重 480/240/120/60mA，默认 180mA（0010），步进 60mA。

【REG04】

Bit7:3 VREG[4:0]：充电稳压电压，相对 3.848V 的偏置，位权重 512/256/128/64/32mV。范围 3.848V~4.616V（11000），默认 4.2V（01011）；高于 11000 的编码钳位到 4.616V。

Bit2:1 TOPOFF_TIMER[1:0]：00=禁用（默认）；01=15min；10=30min；11=45min。表示达到终止条件后的延长充电时间；禁用时满足终止条件即结束充电。

Bit0 VRECHG：再充电阈值。0=120mV（默认），1=240mV。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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Table 8: REG03

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	IPRECHG[3]	0	R/W	by REG_RST by Watchdog	480mA	Precharge Current Default: 180 mA (0010) Offset: 60 mA Note: $I_{PRECHG} > 780$ mA clamped to 780 mA (1100)
6	IPRECHG[2]	0	R/W	by REG_RST by Watchdog	240mA	
5	IPRECHG[1]	1	R/W	by REG_RST by Watchdog	120mA	
4	IPRECHG[0]	0	R/W	by REG_RST by Watchdog	60mA	
3	ITERM[3]	0	R/W	by REG_RST by Watchdog	480mA	Termination Current Default: 180 mA (0010) Offset: 60 mA
2	ITERM[2]	0	R/W	by REG_RST by Watchdog	240mA	
1	ITERM[1]	1	R/W	by REG_RST by Watchdog	120mA	
0	ITERM[0]	0	R/W	by REG_RST by Watchdog	60mA	

Table 9: REG04

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	VREG[4]	0	R/W	by REG_RST by Watchdog	512mV	Charge Voltage Offset: 3.848 V Range: 3.848 V to 4.616 V (11000) Default: 4.2 V (01011) Note: Value above 11000 is clamped to register value 11000 (4.616 V)
6	VREG[3]	1	R/W	by REG_RST by Watchdog	256mV	
5	VREG[2]	0	R/W	by REG_RST by Watchdog	128mV	
4	VREG[1]	1	R/W	by REG_RST by Watchdog	64mV	
3	VREG[0]	1	R/W	by REG_RST by Watchdog	32mV	
2	TOPOFF_TIME R[1]	0	R/W	by REG_RST by Watchdog	00 – Disabled (Default)	The extended time following the termination condition is met. When disabled, charge terminated when termination conditions are met.
1	TOPOFF_TIME R[0]	0	R/W	by REG_RST by Watchdog	01 – 15 minutes 10 – 30 minutes 11 – 45 minutes	
0	VRECHG	0	R/W	by REG_RST by Watchdog	0 – 120 mV 1 – 240 mV	Recharge threshold Default: 120mV

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【REG05】

Bit7 EN_TERM：0=禁用终止，1=使能终止（默认）。

Bit6 Reserved：保留。

Bit5:4 WATCHDOG[1:0]：00=禁用；01=40s（默认）；10=80s；11=160s。

Bit3 EN_TIMER：0=关闭；1=使能快充与预充安全计时器（默认）。

Bit2 CHG_TIMER：0=5h，1=10h（默认）。

Bit1 TREG：热调节阈值，0=90 °C，1=110 °C（默认）。

Bit0 JEITA_ISET（0 °C~10 °C）：0=IC_{CHG} 的 50%；1=IC_{CHG} 的 20%（默认）。

【REG06】

Bit7:6 OVP[1:0]：VBUS OVP 阈值。00=5.5V；01=6.5V（默认，适合 5V 输入）；10=10.5V（9V 输入）；11=14V（12V 输入）。

Bit5:4 BOOSTV[1:0]：Boost 调节电压。00=4.85V；01=5.00V；10=5.15V（默认）；11=5.30V。

Bit3:0 VINDPM[3:0]：绝对 VINDPM 阈值，以 3.9V 为基准，位权重 800/400/200/100mV；范围 3.9V（0000）~5.4V（1111），默认 4.5V（0110）。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



ETA6963

Table 10: REG05

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	EN_TERM	1	R/W	by REG_RST by Watchdog	0 – Disable 1 – Enable	Default: Enable termination (1)
6	Reserved	0	R/W	by REG_RST by Watchdog	Reserved	Reserved
5	WATCHDOG[1]	0	R/W	by REG_RST by Watchdog	00 – Disable timer 01 – 40s	Default: 40s (01)
4	WATCHDOG[0]	1	R/W	by REG_RST by Watchdog	10 – 80s 11 – 160s	
3	EN_TIMER	1	R/W	by REG_RST by Watchdog	0 – Disable 1 – Enable both fast charge and pre- charge timer	Default: Enable (1)
2	CHG_TIMER	1	R/W	by REG_RST by Watchdog	0 – 5 hrs 1 – 10 hrs	Default: 10 hours (1)
1	TREG	1	R/W	by REG_RST by Watchdog	Thermal Regulation Threshold: 0 - 90°C 1 - 110°C	Default: 110°C (1)
0	JEITA_ISET (0°C-10°C)	1	R/W	by REG_RST by Watchdog	0 – 50% of ICHG 1 – 20% of ICHG	Default: 20% (1)

Table 11: REG06

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	OVP[1]	0	R/W	by REG_RST	Default: 6.5V (01)	V _{BUS_OVP} threshold: 00 - 5.5 V 01 – 6.5 V (5V input) 10 – 10.5 V (9V input) 11 – 14 V (12V input)
6	OVP[0]	1	R/W	by REG_RST		
5	BOOSTV[1]	1	R/W	by REG_RST	Default: 5.15V (10)	Boost Regulation Voltage: 00 - 4.85V 01 - 5.00V 10 - 5.15V 11 - 5.30V
4	BOOSTV[0]	0	R/W	by REG_RST		
3	VINDPM[3]	0	R/W	by REG_RST	800mV	Absolute VINDPM Threshold Offset: 3.9 V Range: 3.9 V (0000) – 5.4 V (1111) Default: 4.5V (0110)
2	VINDPM[2]	1	R/W	by REG_RST	400mV	
1	VINDPM[1]	1	R/W	by REG_RST	200mV	
0	VINDPM[0]	0	R/W	by REG_RST	100mV	

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【REG07】

Bit7 IINDET_EN：0=不强制输入限流检测；1=在 VBUS 存在时强制执行输入限流检测。检测完成后自动返回 0。

Bit6 TMR2X_EN：0=关闭；1=当处于输入 DPM（电压或电流）、JEITA 冷温区或热调节时，使安全计时器以 2 倍时间/半速计数。

Bit5 BATFET_DIS：0=允许 Q4 导通（默认）；1=在 TBATFET_DLY 延迟后关闭 Q4。

Bit4 JEITA_VSET（45 ° C~60 ° C）：0=充电电压设置为最大 4.1V；1=设置为 VREG。

Bit3 BATFET_DLY：0=设置 BATFET_DIS 后立即关闭；1=延迟 TBATFET_DLY（典型 10s）后关闭，默认 1。

Bit2 BATFET_RST_EN：0=禁用 BATFET 复位；1=使能 BATFET 复位（默认）。

Bit1:0 VDPM_BAT_TRACK[1:0]：00=关闭，VINDPM 完全由寄存器设定；01=VBAT+200mV；10=VBAT+250mV；11=VBAT+300mV。实际 VINDPM 取寄存器值与 VBAT+跟踪偏移中的较高者。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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Table 12: REG07

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION	NOTE
7	IINDET_EN	0	R/W	by REG_RST by Watchdog	0 - Not in input current limit detection 1 - Force input current limit detection when VBUS is present	Returns to 0 after input detection is complete
6	TMR2X_EN	1	R/W	by REG_RST by Watchdog	0 - Disable 1 - Safety timer slowed by 2X during input DPM (both V and I) or JEITA cool, or thermal regulation	
5	BATFET_DIS	0	R/W	by REG_RST	0 - Allow Q4 turn on, 1 - Turn off Q4 with TBATFET_DLY delay time (REG07[3])	Default: Allow Q4 turn on (0)
4	JEITA_VSET (45°C-60°C)	0	R/W	by REG_RST by Watchdog	0 - Set Charge Voltage to 4.1V (max), 1 - Set Charge Voltage to VREG	
3	BATFET_DLY	1	R/W	by REG_RST	0 - Turn off BATFET immediately when BATFET_DIS bit is set 1 - Turn off BATFET after TBATFET_DLY (typ. 10s) when BATFET_DIS bit is set	Default: 1 Turn off BATFET after T_{BATFET_DLY} (typ. 10s) when BATFET_DIS bit is set
2	BATFET_RST_EN	1	R/W	by REG_RST by Watchdog	0 - Disable BATFET reset function 1 - Enable BATFET reset function	Default: 1 Enable BATFET reset function
1	VDPM_BAT_TRACK[1]	0	R/W	by REG_RST	00 - Disable function (VINDPM set by register)	Sets VINDPM to track BAT voltage. Actual VINDPM is higher of register value and $V_{BAT} + V_{OPM_BAT_TRACK}$
0	VDPM_BAT_TRACK[0]	0	R/W	by REG_RST	01 - $V_{BAT} + 200mV$ 10 - $V_{BAT} + 250mV$ 11 - $V_{BAT} + 300mV$	

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【REG08 - 状态寄存器】

Bit7:5 VBUS_STAT[2:0]：000=无输入；001=USB Host SDP；010=USB CDP（1.5A）；011=USB DCP（2.4A）；101=未知适配器（500mA）；110=非标准适配器（1A/2A/2.1A/2.4A）；111=OTG。软件限流值从 IINDPM 读取。

Bit4:3 CHRG_STAT[1:0]：00=未充电；01=预充（VBAT<VBATLOW）；10=快充；11=充电终止。

Bit2 PG_STAT：0=Power Not Good；1=Power Good。

Bit1 THERM_STAT：0=未处于热调节；1=处于热调节。

Bit0 VSYS_STAT：0=未处于 VSYS_MIN 调节（VBAT>VSYS_MIN）；1=处于 VSYS_MIN 调节（VBAT<VSYS_MIN）。

【REG09 - 故障寄存器】

Bit7 WATCHDOG_FAULT：0=正常；1=看门狗超时。

Bit6 BOOST_FAULT：0=正常；1=OTG 时 VBUS 过载、VBUS_OVP、电池过低或其他无法启动 Boost 的条件。

Bit5:4 CHRG_FAULT[1:0]：00=正常；01=输入故障（VBUS_OVP 或 VBAT<VBUS<3.8V）；10=热关断；11=充电安全计时器超时。

Bit3 BAT_FAULT：0=正常；1=BATOVP。

Bit2:0 NTC_FAULT[2:0]：Buck/JEITA 模式下 000=正常、010=Warm、011=Cool、101=Cold、110=Hot；Boost 模式下 000=正常、101=Cold、110=Hot。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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Table 13: REG08

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION
7	VBUS_STAT[2]	X	R	NA	VBUS Status register 000: No input 001: USB Host SDP 010: USB CDP: (1.5A) 011: USB DCP (2.4 A) 101: Unknown Adapter (500 mA) 110: Non-Standard Adapter (1A/2A/2.1A/2.4A) 111: OTG Software current limit is reported in IINDPM register
6	VBUS_STAT[1]	X	R	NA	
5	VBUS_STAT[0]	X	R	NA	
4	CHRG_STAT[1]	X	R	NA	
3	CHRG_STAT[0]	X	R	NA	
2	PG_STAT	X	R	NA	Power Good status: 0 – Power Not Good 1 – Power Good
1	THERM_STAT	X	R	NA	0 – Not in thermal regulation 1 – in thermal regulation
0	VSYS_STAT	X	R	NA	0 – Not in V_{SYS_MIN} regulation ($V_{BAT} > V_{SYS_MIN}$) 1 – in V_{SYS_MIN} regulation ($V_{BAT} < V_{SYS_MIN}$)

Table 14: REG09

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION
7	WATCHDOG_FAULT	X	R	NA	0 – Normal, 1- Watchdog timer expiration
6	BOOST_FAULT	X	R	NA	0 – Normal, 1 – VBUS overloaded in OTG, or VBUS_OVP, or battery is too low (any conditions that we cannot start boost function)
5	CHRG_FAULT[1]	X	R	NA	00 – Normal, 01 – input fault (V_{BUS_OVP} or $V_{BAT} < V_{BUS} < 3.8$ V)
4	CHRG_FAULT[0]	X	R	NA	10 -Thermal shutdown, 11 – Charge Safety Timer Expiration
3	BAT_FAULT	X	R	NA	0 – Normal, 1 – BATOV
2	NTC_FAULT[2]	X	R	NA	JEITA 000 – Normal, 010 – Warm, 011 – Cool, 101 – Cold, 110 – Hot (Buck mode) 000 – Normal, 101 – Cold, 110 – Hot (Boost mode)
1	NTC_FAULT[1]	X	R	NA	
0	NTC_FAULT[0]	X	R	NA	

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【REG0A】

Bit7 VBUS_GD：0=未接入 VBUS；1=已接入 VBUS。
Bit6 VINDPM_STAT：0=未处于 VINDPM；1=处于 VINDPM。
Bit5 IINDPM_STAT：0=未处于 IINDPM；1=处于 IINDPM。
Bit4 Reserved：保留。
Bit3 TOPOFF_ACTIVE：0=Top-Off 计时器未计数；1=正在计数。
Bit2 ACOV_STAT：0=器件未处于 ACOV；1=器件处于 ACOV。
Bit1 VINDPM_INT_MASK：0=允许 VINDPM INT 脉冲；1=屏蔽 VINDPM INT。
Bit0（原文名称重复为 VINDPM_INT_MASK，语义为 IINDPM_INT_MASK）：0=允许 IINDPM INT 脉冲；1=屏蔽 IINDPM INT。

【REG0B】

Bit7 REG_RST：0=保持当前寄存器设置；1=恢复默认寄存器值并复位安全计时器；复位完成后该位自动回 0。
Bit6:3 PIN[3:0]：ETA6963 固定识别码 0111。
Bit2 ETA_PART_ID：ETA 器件识别位。
Bit1:0 DEV_REV[1:0]：器件修订版本。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。



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Table 15: REG0A

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION
7	VBUS_GD	X	R	NA	0 – Not V _{BUS} attached 1 – V _{BUS} Attached
6	VINDPM_STAT	X	R	NA	0 – Not in VINDPM 1 – in VINDPM
5	IINDPM_STAT	X	R	NA	0 – Not in IINDPM 1 – in IINDPM
4	Reserved	X	R	NA	Reserved Reserved
3	TOPOFF_ACTIVE	X	R	NA	0 – Top off timer not counting 1 – Top off timer counting
2	ACOV_STAT	X	R	NA	0 – Device is NOT in ACOV 1 – Device is in ACOV
1	VINDPM_INT_MASK	0	R/W	by REG_RST	0 - Allow VINDPM INT pulse 1 - Mask VINDPM INT pulse
0	IINDPM_INT_MASK	0	R/W	by REG_RST	0 - Allow IINDPM INT pulse 1 - Mask IINDPM INT pulse

Table 16: REG0B

BIT	NAME	POR	TYPE	RESET BY	DESCRIPTION
7	REG_RST	X	R/W	NA	Register reset 0 – Keep current register setting 1 – Reset to default register value and reset safety timer Note: Bit resets to 0 after register reset is completed
6	PIN[3]	0	R	NA	ETA6963: 0111
5	PIN[2]	1	R	NA	
4	PIN[1]	1	R	NA	
3	PIN[0]	1	R	NA	
2	ETA_PART ID	1	R	NA	ETA Part ID for recognition
1	DEV_REV[1]	X	R	NA	
0	DEV_REV[0]	X	R	NA	

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【功能方框图】

方框图展示了 ETA6963 的完整内部架构，包括：BUS 输入与 Q1-BRFET、PMID、Q2-HSFET/Q3-LSFET 同步 Buck/Boost 功率级、BST/LDO、Q4-BATFET、SYS/BAT 电流检测、VBUS/ACOV/UVLO、VINDPM/IINDPM、充电电压/电流环、PWM/PFM 控制、数字控制、充电控制、热保护、过放/过流保护、TS 热敏电阻检测、DP/DM 输入源检测、I2C 接口、nINT/STAT 等。信号名、比较器名和寄存器位名保持原文，具体连接见原页方框图。

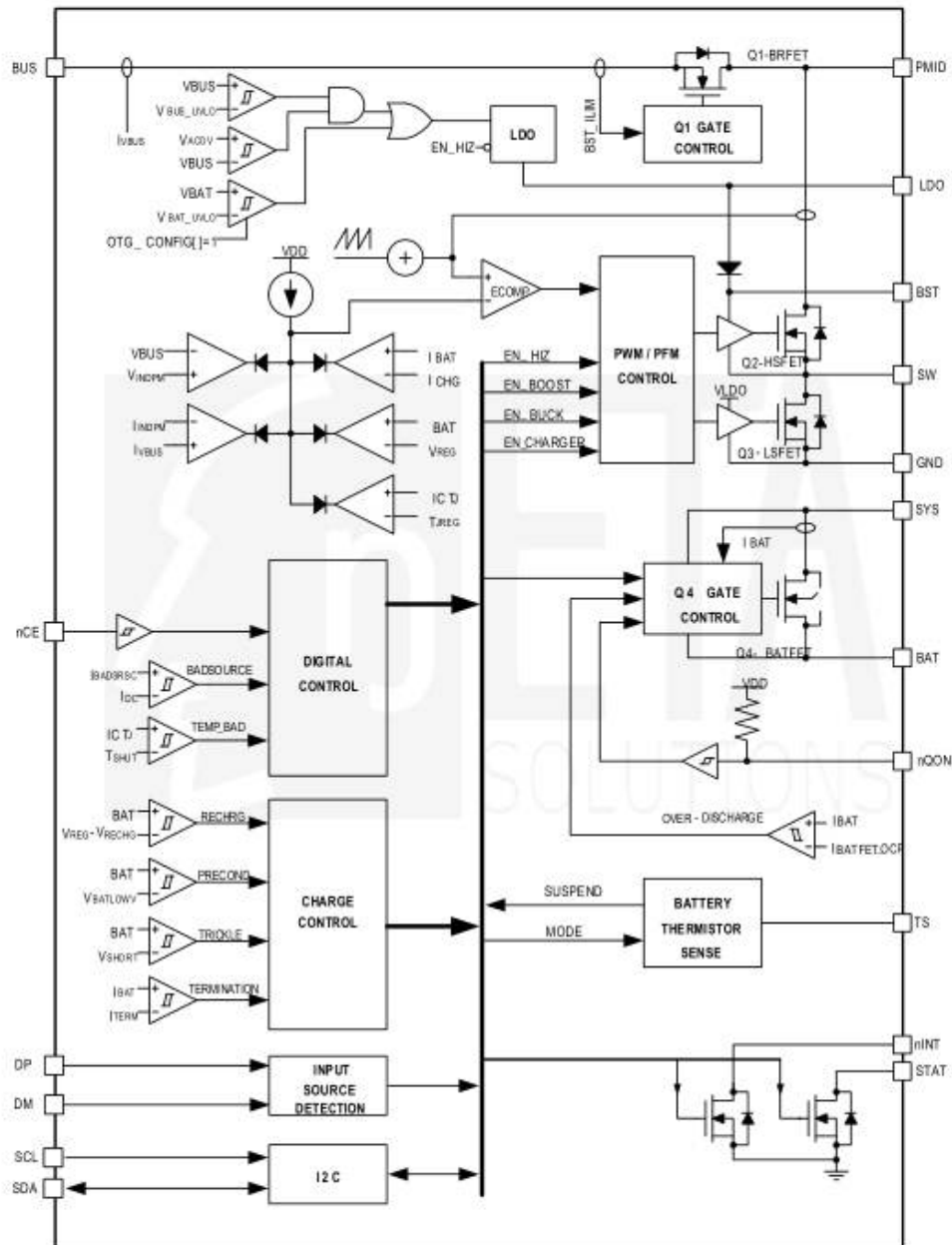
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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FUNCTION BLOCK DIAGRAM



原文第 31 页 - 中文译文

【PCB 设计指南】

为尽可能获得干净的转换器供电，应遵循以下建议：

1. BUS、PMID、BAT、SYS、LDO 的电容尽可能靠近相应引脚；PGND 在底层使用宽铜连接，并增加尽可能多的 PGND 过孔，减小 PMID 电容回路的铜阻。
2. 电感输入端尽可能靠近 SW。该开关节点铜面积应尽量小以降低电场/磁场辐射，但走线必须足够宽以承载充电电流。不要通过多层并联走线连接 SW-电感，并尽量降低该区域与其他走线或平面的寄生电容。
3. 输出电容靠近电感和器件放置，其地端通过短铜线或 GND 平面连接到 IC 地。
4. 将裸露散热焊盘作为单一接地点。
5. 充电器功率地与模拟地采用单点连接，连接点尽量位于器件正下方。可使用铺地铜，但应避免功率引脚附近以降低电感/电容耦合噪声。
6. 去耦电容紧邻 IC 引脚，走线尽可能短。
7. 封装背面的裸露散热焊盘必须焊接到 PCB 地，并在 IC 正下方设置足够热过孔连接其他层地平面。
8. 过孔数量和尺寸应保证目标电流路径具有足够铜截面积。
9. 除必要的桥接连接外，底层应尽可能大面积铺地。

原页给出了推荐器件、电容、电感及地连接布局示意。

原页图示参考

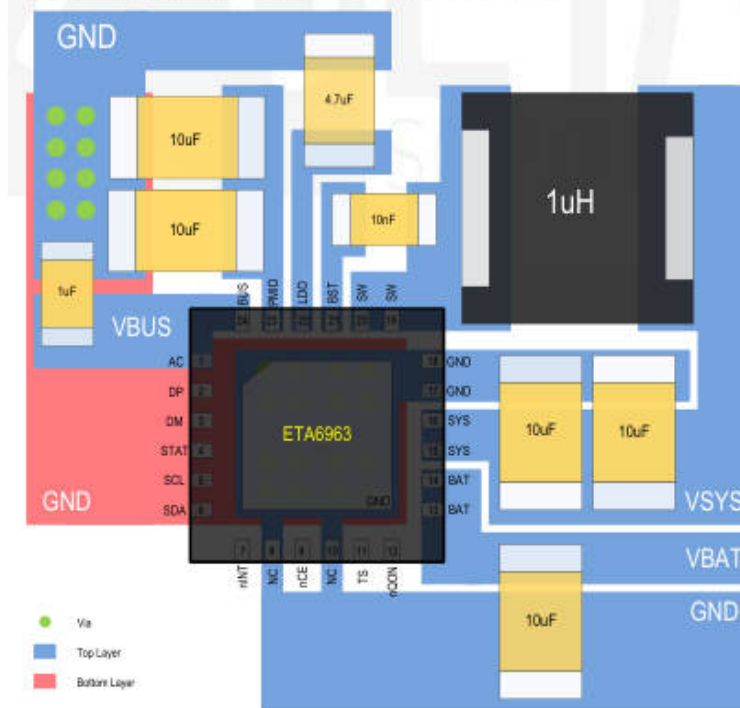
以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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PCB DESIGN GUIDELINE

In order to have as clean as possible supply for converter, please follow following suggestion:

1. Place the BUS, PMID, BAT, SYS, LDO capacitor as close as possible to the pins and wide bottom layer for PGND connections. Also add as much as possible vias for PGND to minimize copper resistance added to PMID capacitor.
2. Place inductor input pin to SW pin as close as possible. Minimize the copper area of this trace to lower electrical and magnetic field radiation but make the trace wide enough to carry the charging current. Do not use multiple layers in parallel for this connection. Minimize parasitic capacitance from this area to any other trace or plane.
3. Put output capacitor near to the inductor and the device. Ground connections need to be tied to the IC ground with a short copper trace connection or GND plane.
4. Use thermal pad as the single ground connection point.
5. Use single ground connection to tie charger power ground to charger analog ground. Just beneath the device. Use ground copper pour but avoid power pins to reduce inductive and capacitive noise coupling.
6. Place decoupling capacitors next to the IC pins and make trace connection as short as possible.
7. It is critical that the exposed thermal pad on the backside of the device package be soldered to the PCB ground. Ensure that there are sufficient thermal vias directly under the IC, connecting to the ground plane on the other layers.
8. Ensure that the number and sizes of vias allow enough copper for a given current path.
9. Ensure almost the bottom layer be ground unless bridge connection.



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ETA6963_V1.7

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原文第 32 页 - 中文译文

【封装外形】

QFN4×4-24。原页分别给出了两个封装/组装厂版本的机械尺寸图。尺寸、焊盘、引脚编号及公差请以原页图示为准。

原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

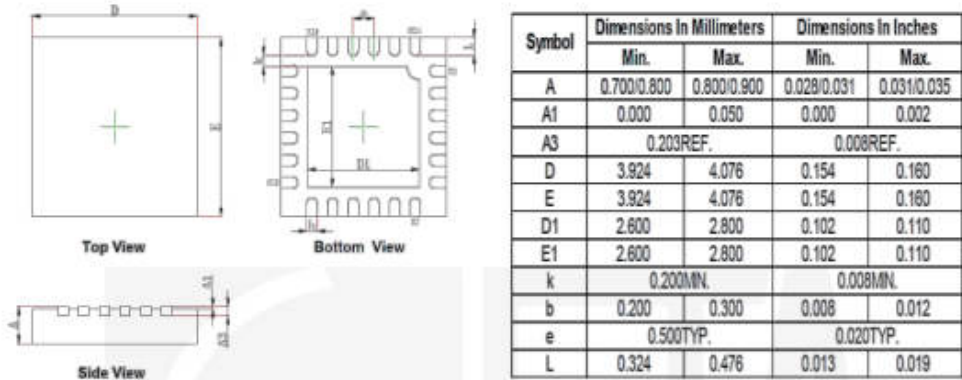


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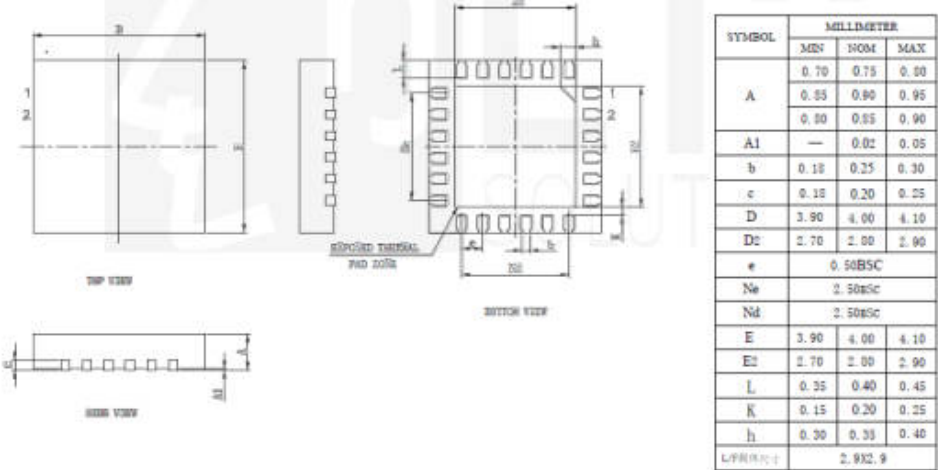
PACKAGE OUTLINE

QFN4x4-24

From assembly house 1:



From assembly house 2:



原文第 33 页 - 中文译文

【卷带包装信息】

原页给出了 ETA6963 QFN4×4-24 的 Tape and Reel 卷带包装结构和尺寸信息。卷带方向、定位孔、器件方向及尺寸请参照原页图示。

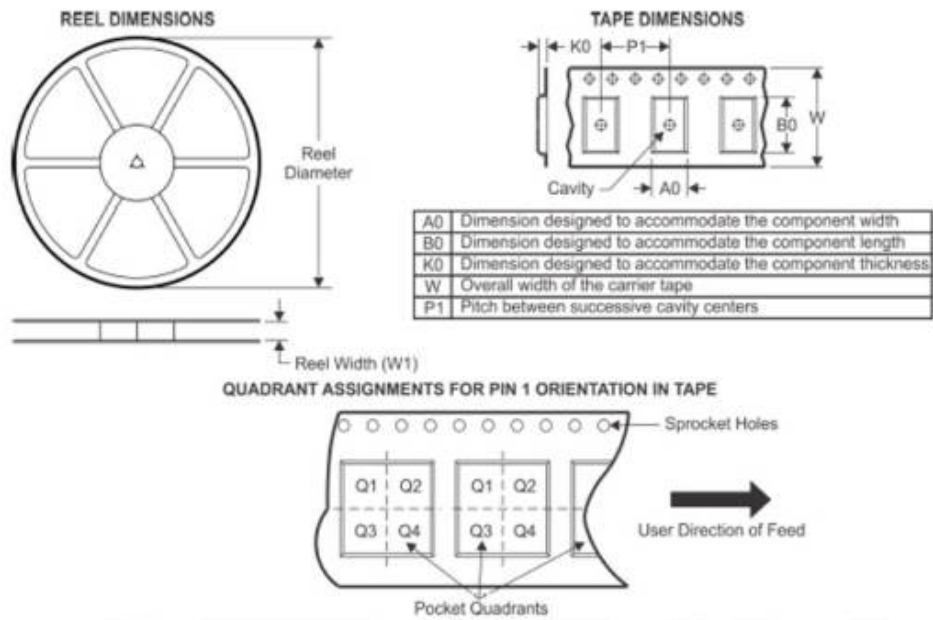
原页图示参考

以下图像保留原始表格、曲线、封装/引脚图和信号标识，用于与中文译文交叉核对。

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TAPE AND REEL INFORMATION



Device	Package Type	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
ETA6963Q4Y	QFN4-24	24	5000	330	12.4	4.3	4.3	1.1	8	12	Q2